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Fitzpatrick et al.

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(54) **SYSTEM AND METHOD FOR
MULTILATERAL GUNSHOT DETECTION**

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U.S.C. 154(b) by 0 days.

2010/0245582 A1 9/2010 Harel
2016/0241818 A1 8/2016 Palanisamy et al.
2017/0154638 A1 6/2017 Hwang et al.
2017/0353788 A1 12/2017 Tatematsu et al.
2019/0311604 A1 10/2019 Morehouse et al.
2019/0350065 A1* 11/2019 Stuby, Jr. F21V 23/045
2020/0143649 A1* 5/2020 Coonley G08B 17/08
2020/0217468 A1* 7/2020 Brown H04W 4/90
2020/0381006 A1* 12/2020 Davis G10L 25/24
2021/0110843 A1 4/2021 Davis et al.
2022/0050479 A1* 2/2022 Hurst G05D 1/104
2022/0060663 A1 2/2022 Guibene et al.
2022/0295176 A1 9/2022 Eswara et al.

FOREIGN PATENT DOCUMENTS

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OTHER PUBLICATIONS

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PCT/US2023/035133 dated Nov. 17, 2023 (4 pages).

* cited by examiner

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G08B 23/00 (2006.01)
G08B 25/00 (2006.01)
G10L 21/14 (2013.01)
G10L 25/27 (2013.01)
G10L 25/87 (2013.01)

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(52) **U.S. Cl.**
CPC **G10L 25/87** (2013.01); **G10L 21/14**
(2013.01); **G10L 25/27** (2013.01)

(57) **ABSTRACT**

(58) **Field of Classification Search**
CPC G08B 21/00; G08B 23/00; G08B 25/00
See application file for complete search history.

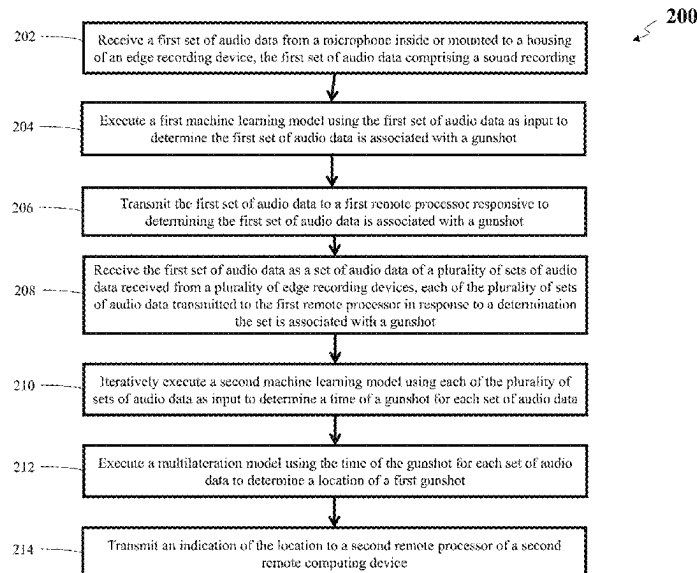
An apparatus for detecting gunshots. The apparatus may include a device housing configured to removably couple to a light fixture; a microphone inside or mounted to the device housing; and a processor inside the device housing and electrically coupled to the microphone. The processor can be configured to receive audio data from the microphone; execute a machine learning model using the audio data as input to determine whether the audio data corresponds to a gunshot; and responsive to determining the audio data corresponds to a gunshot, transmit the audio data to a remote processor.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,424,048 B1* 9/2019 Calhoun G06N 3/04
11,282,353 B1 3/2022 Fowler et al.

19 Claims, 23 Drawing Sheets



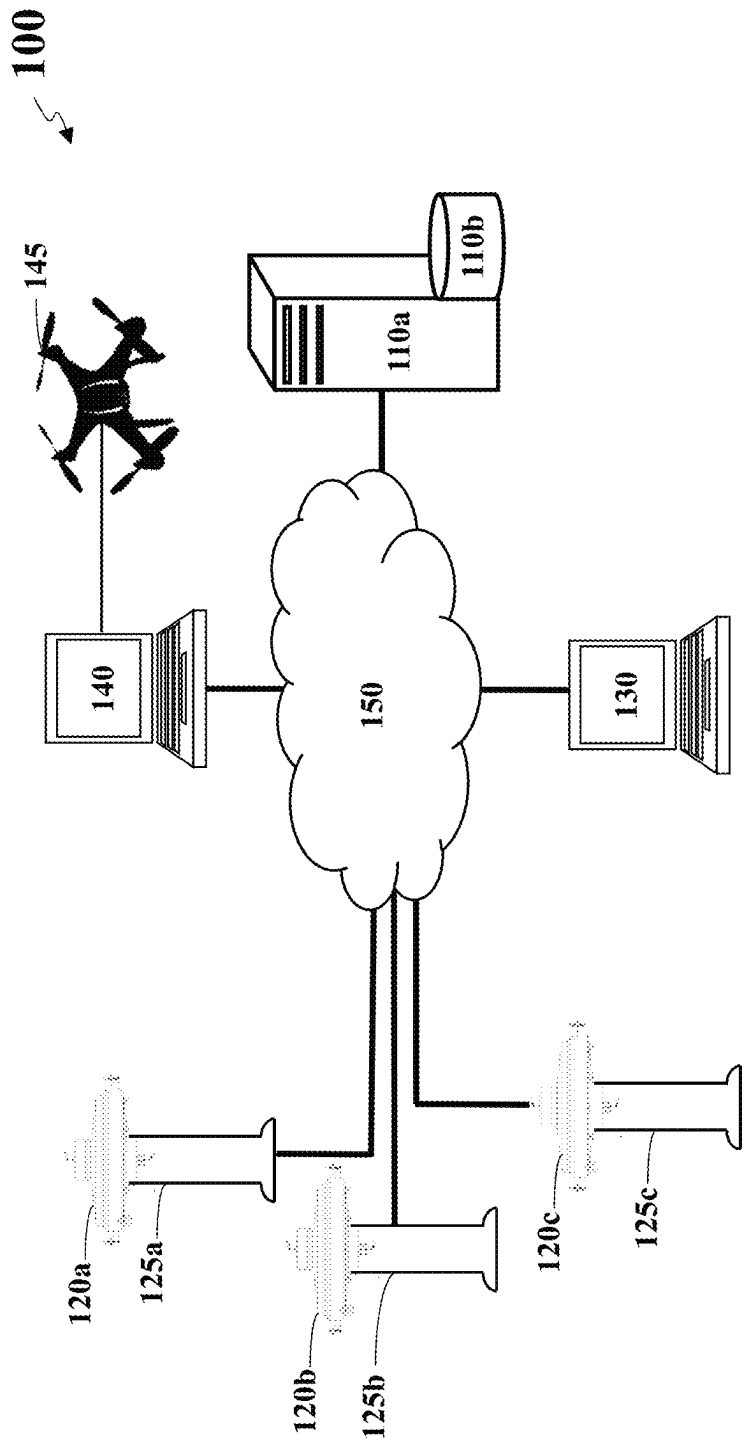


FIG. 1

200

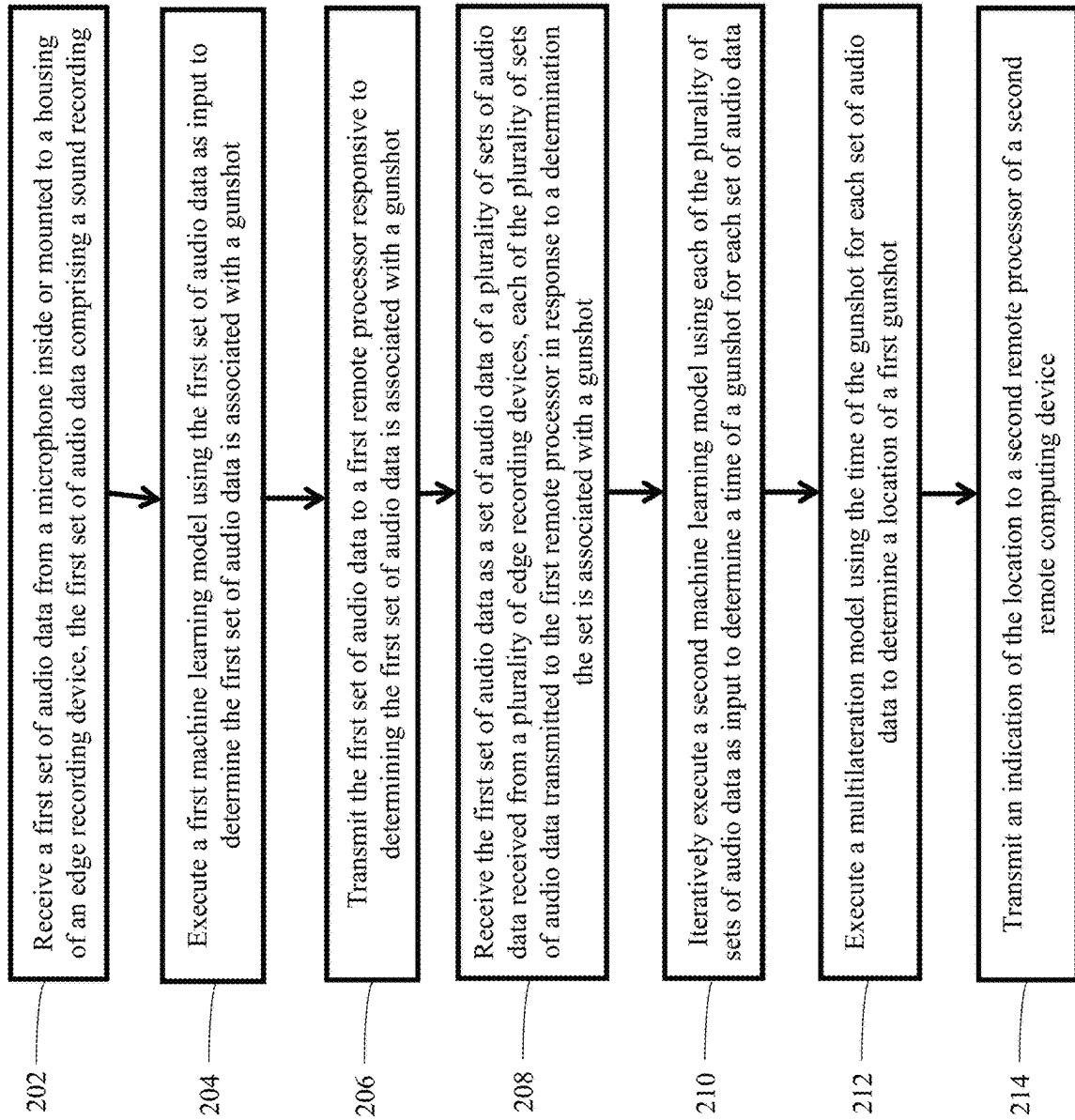


FIG. 2

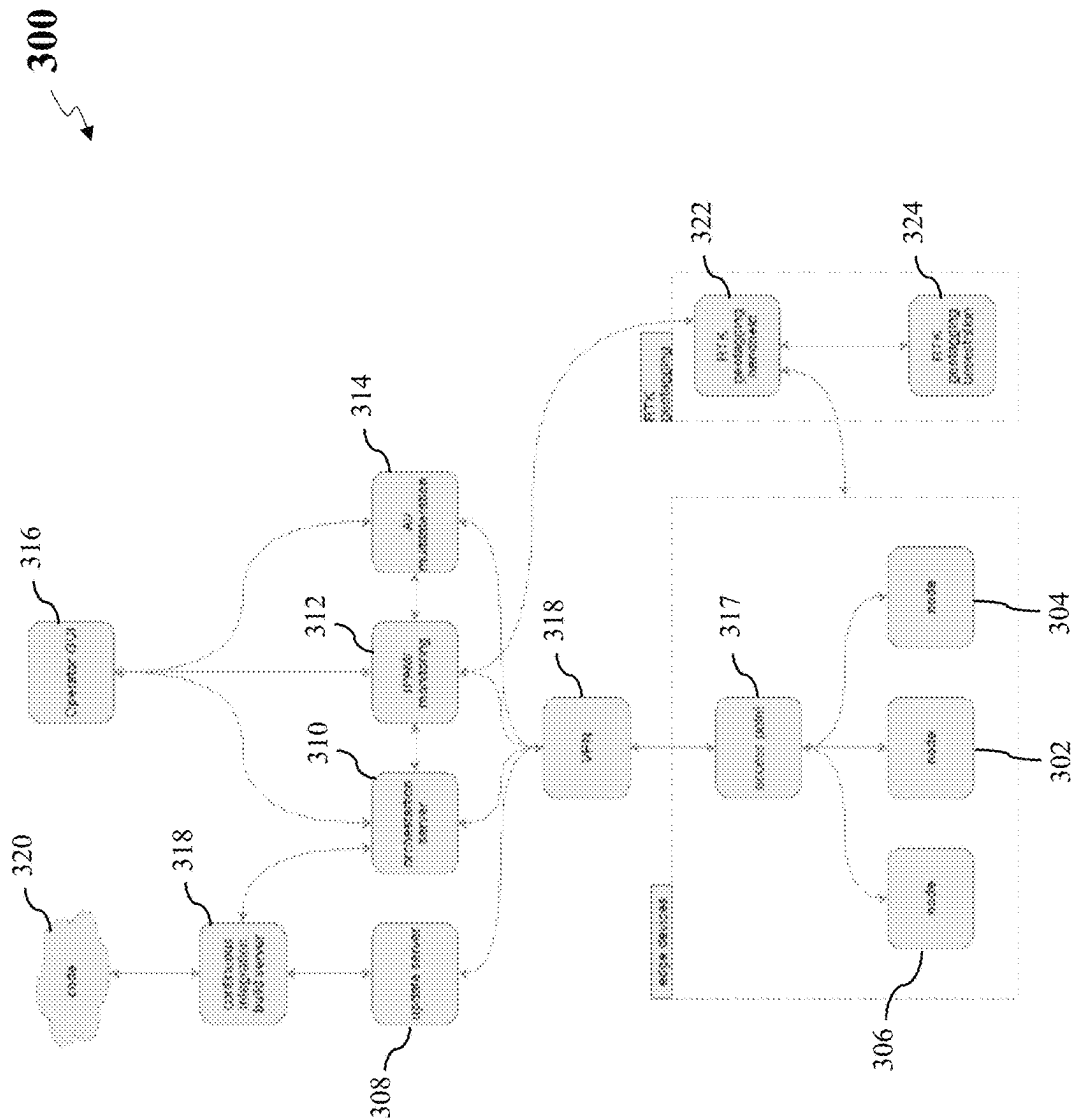


FIG. 3

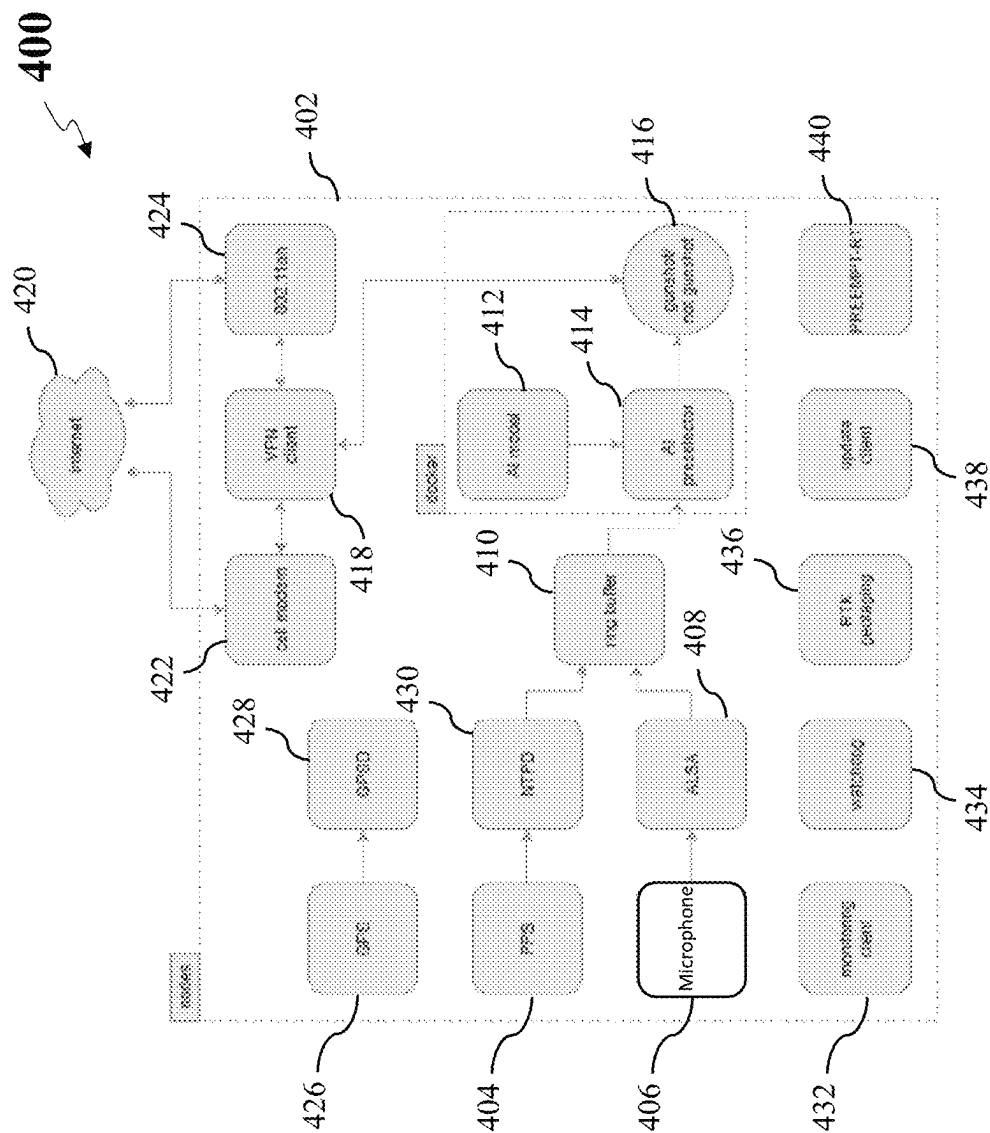


FIG. 4

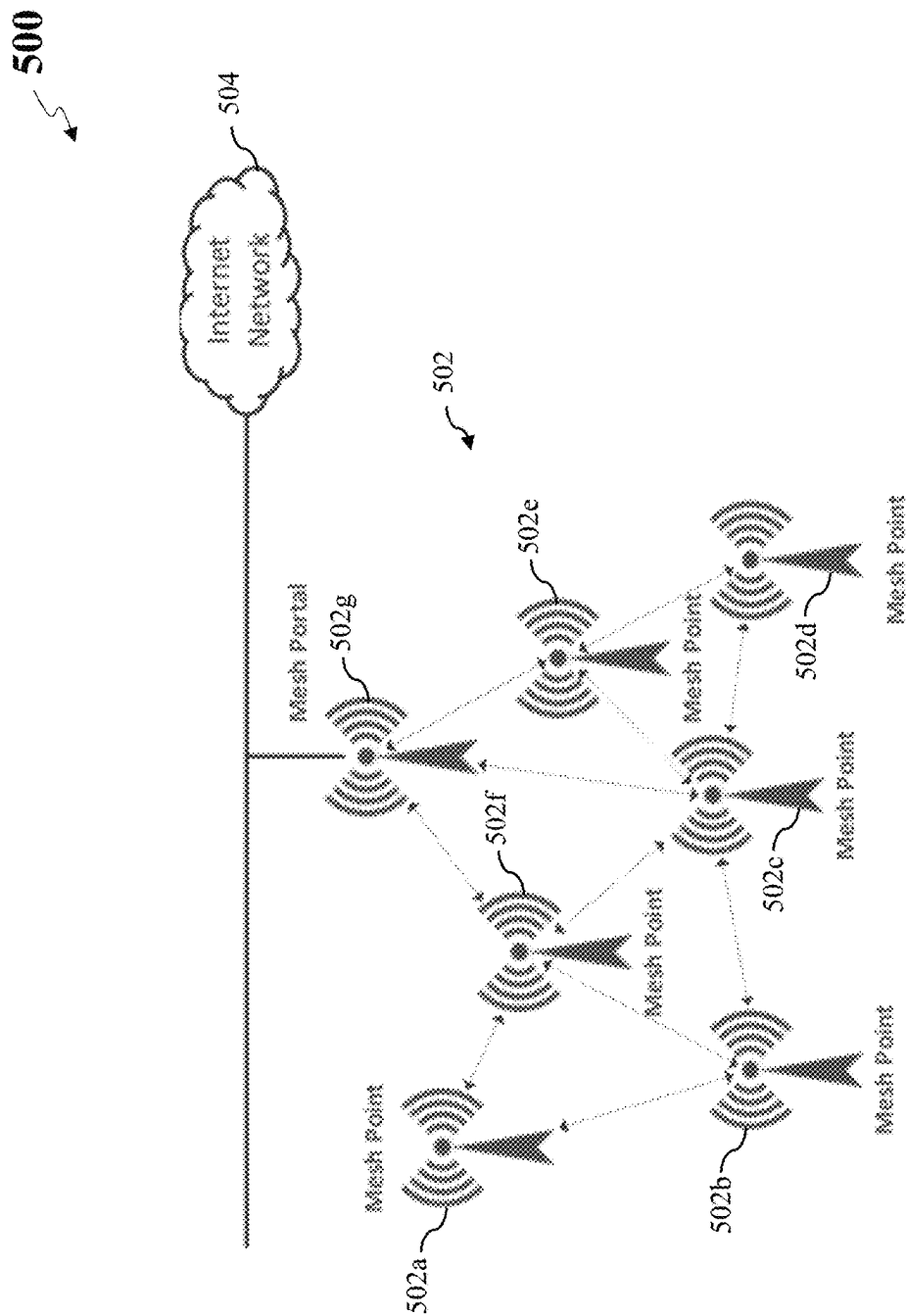


FIG. 5

600

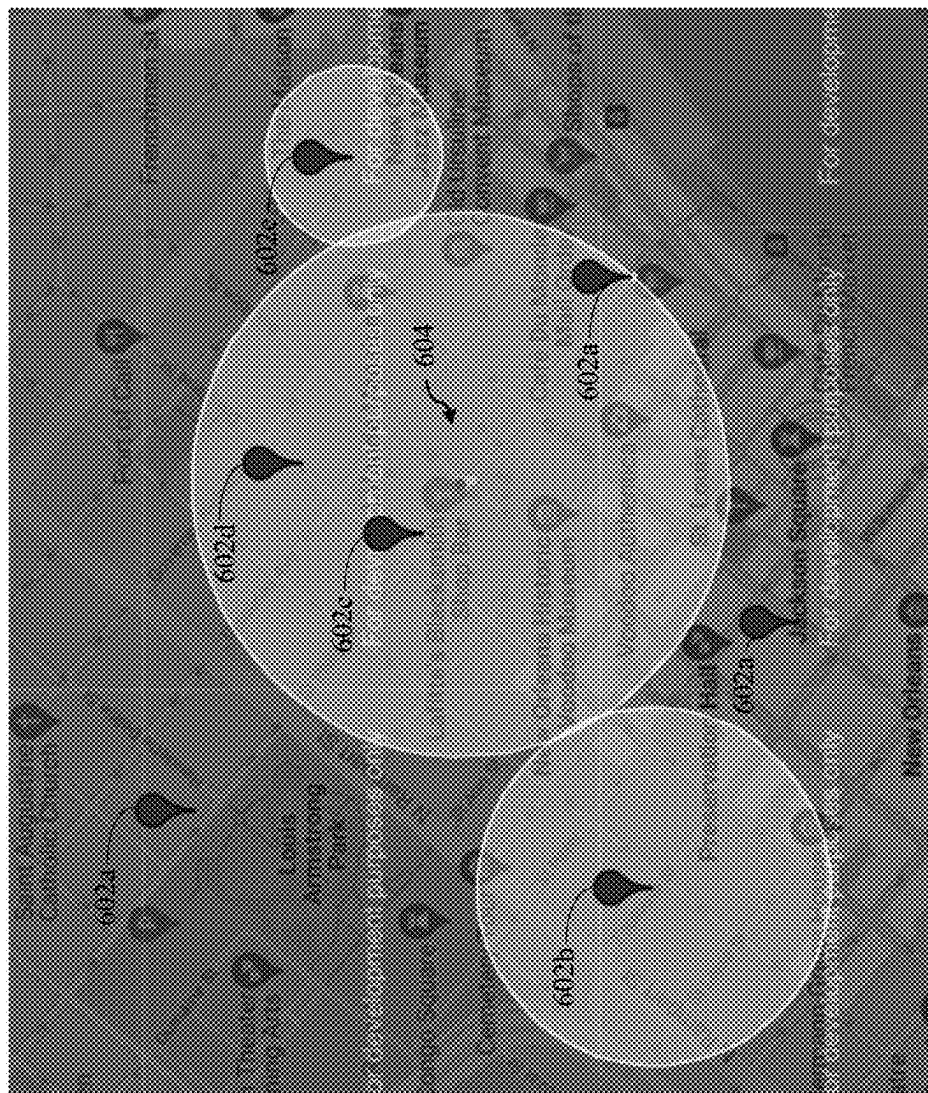


FIG. 6

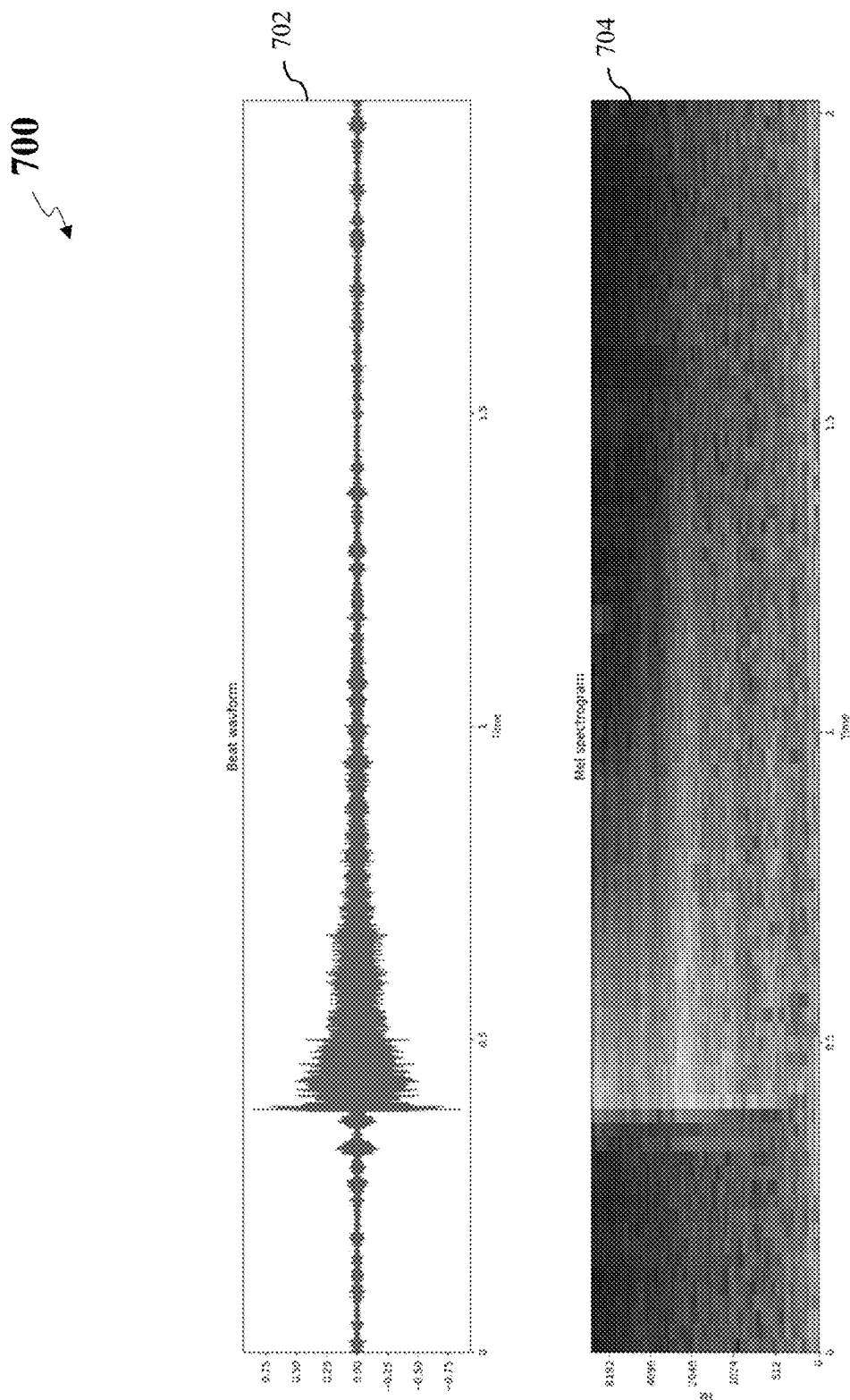


FIG. 7

800

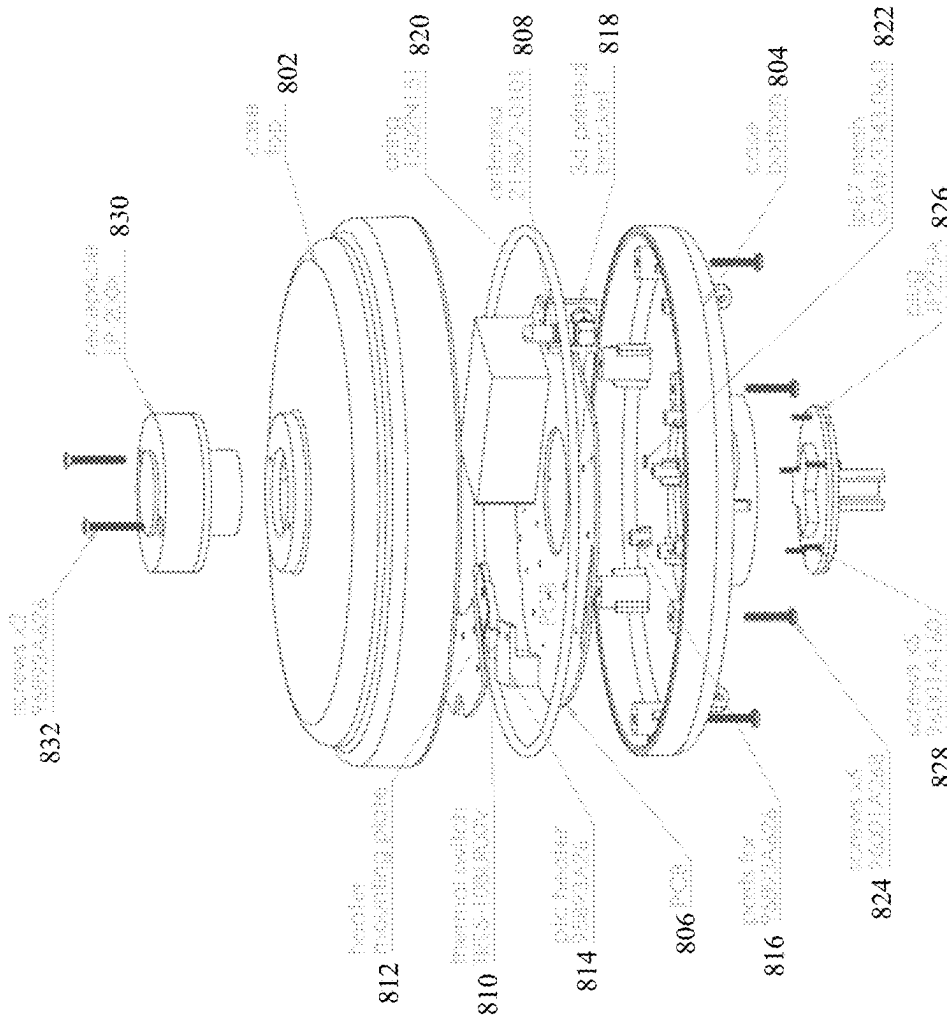


FIG. 8

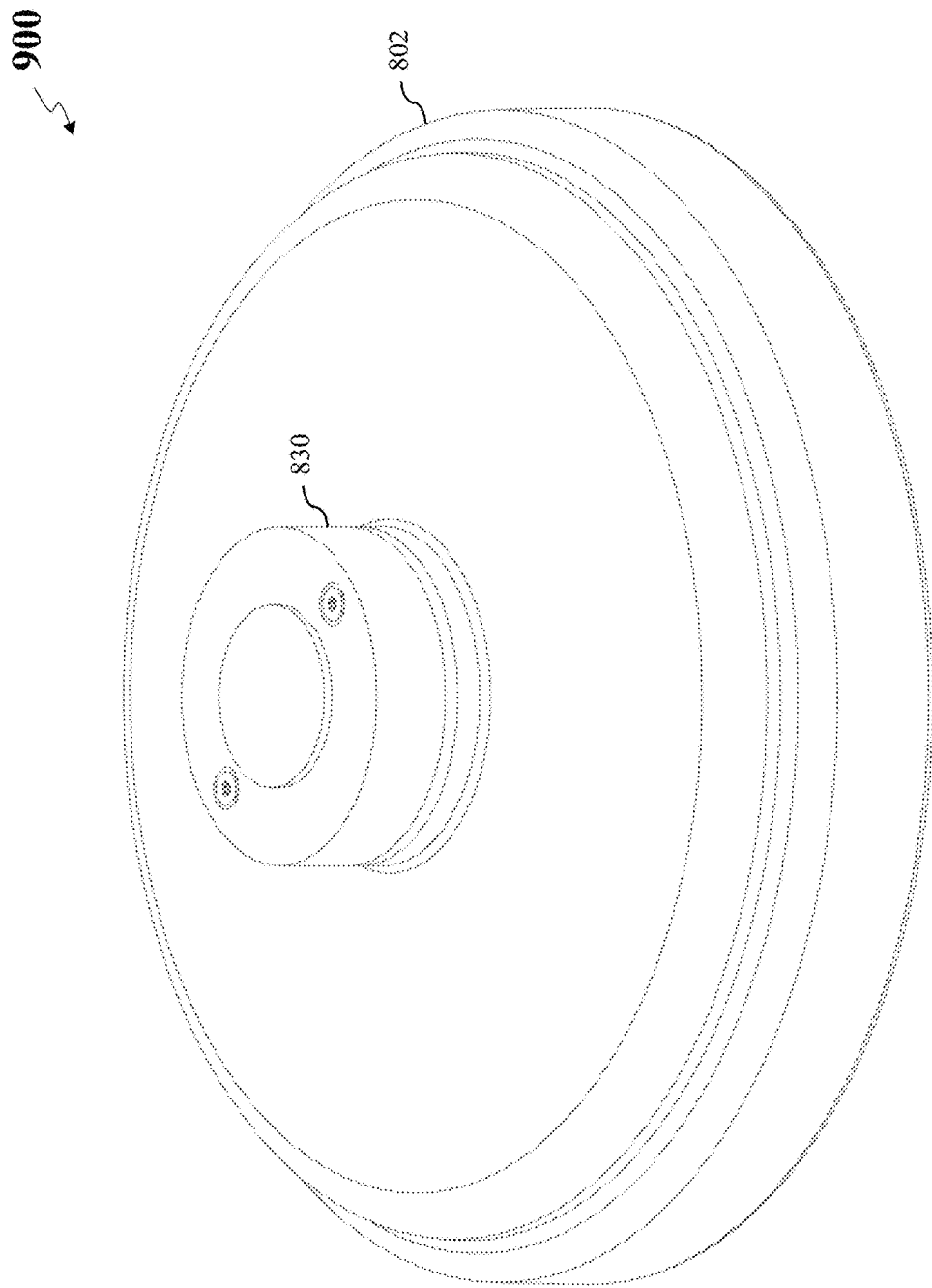


FIG. 9

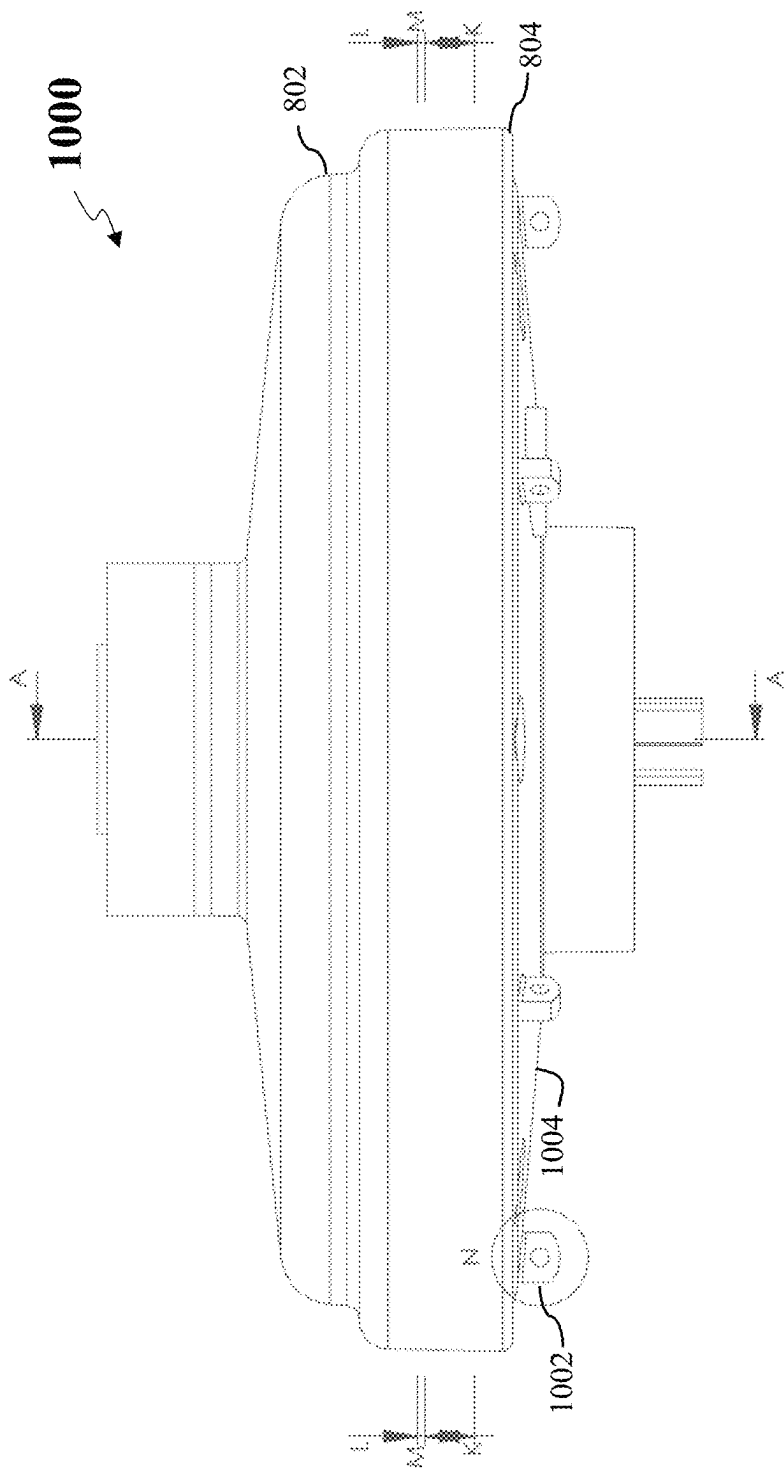


FIG. 10

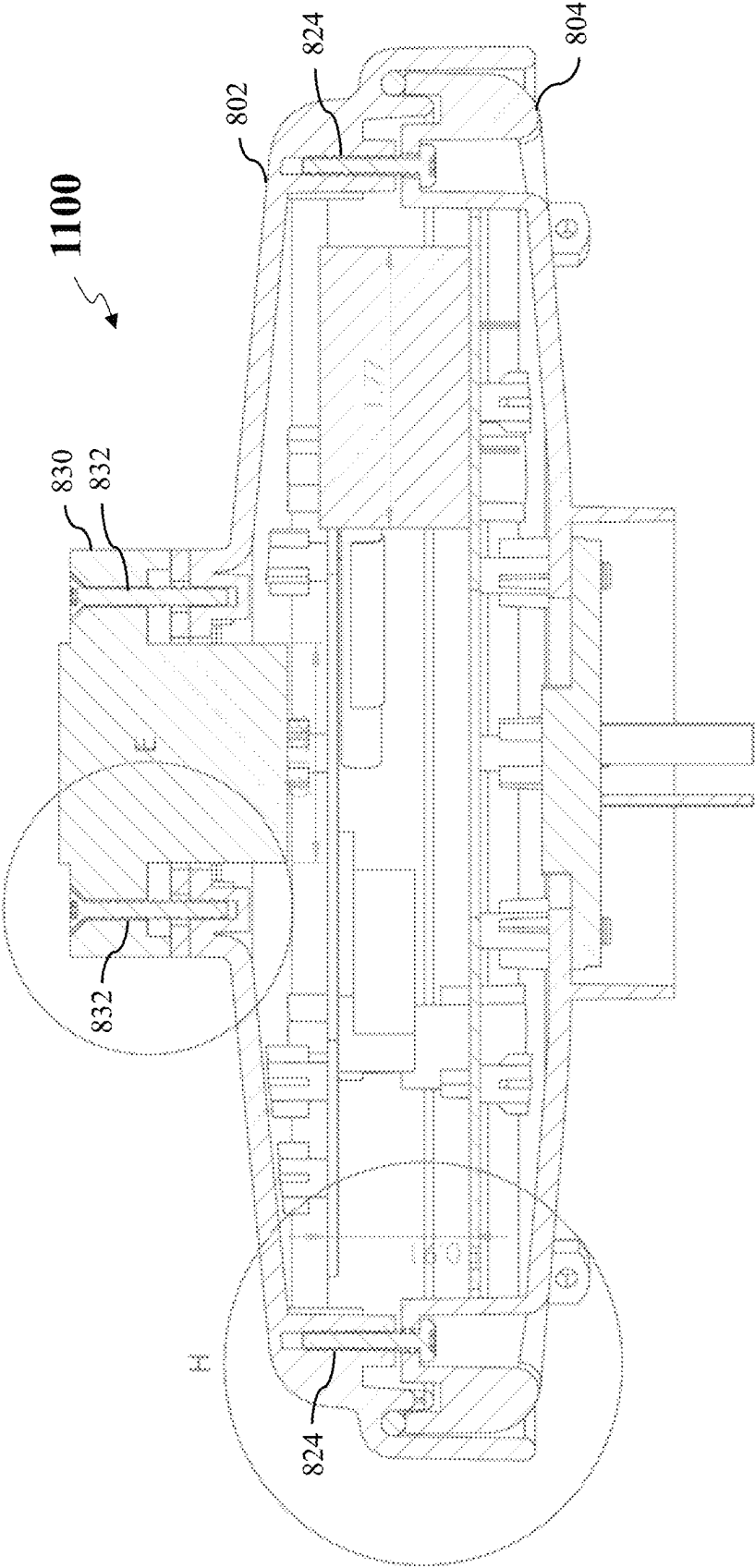


FIG. 11

1200

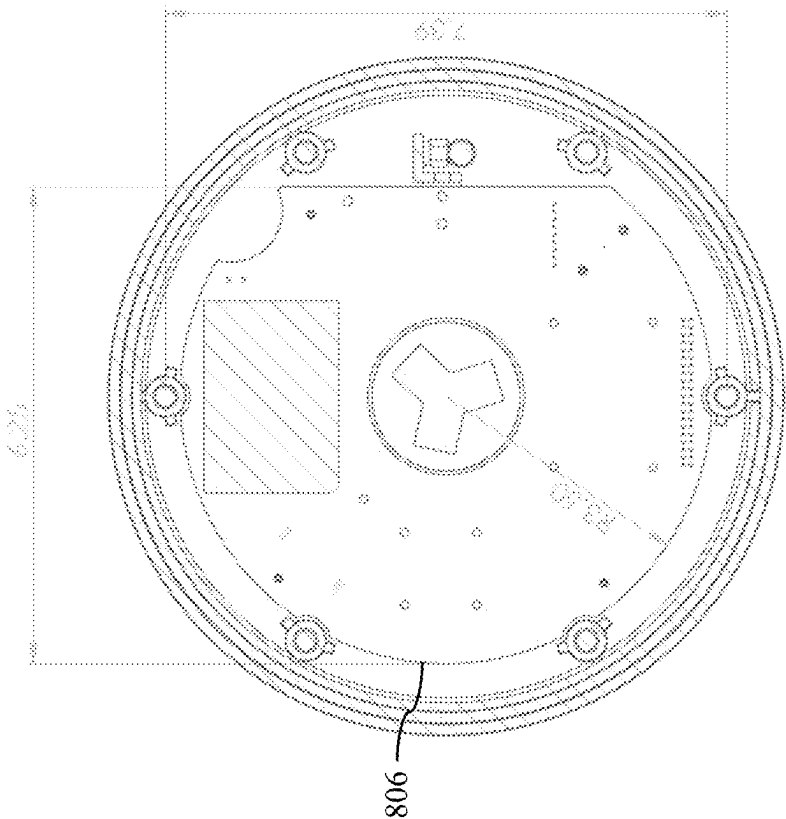


FIG. 12

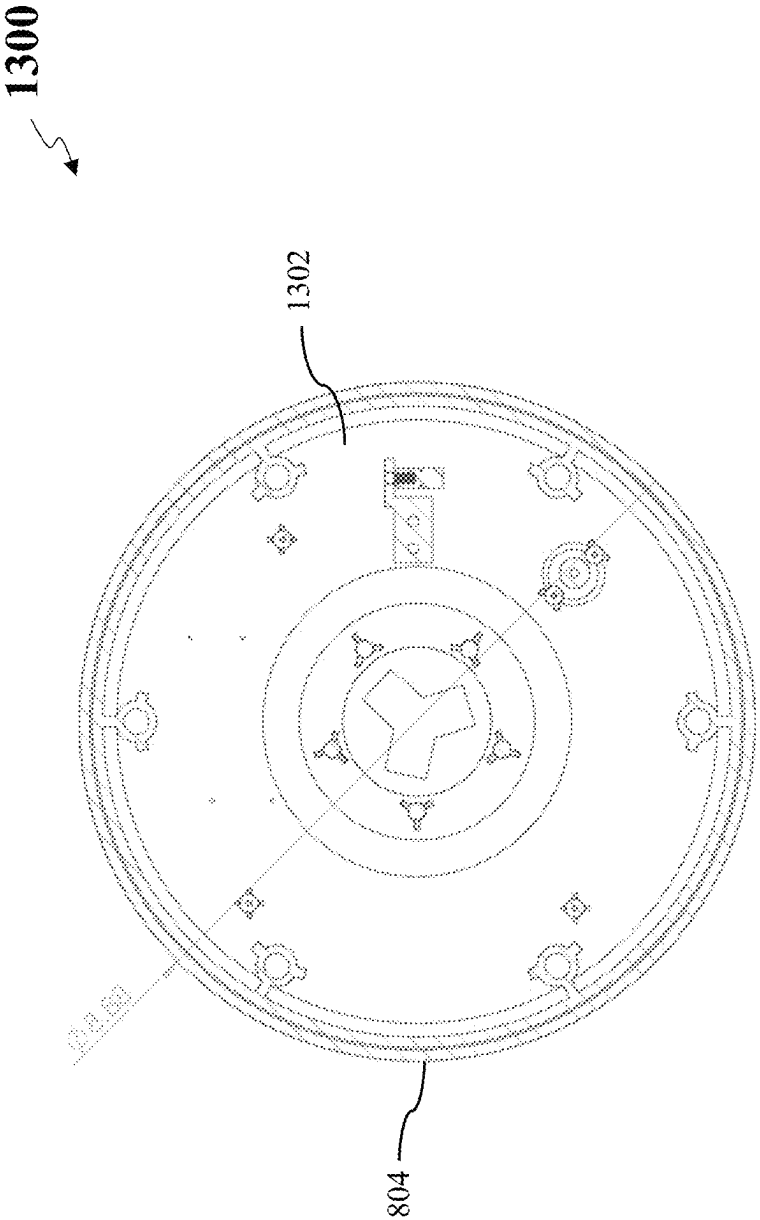


FIG. 13

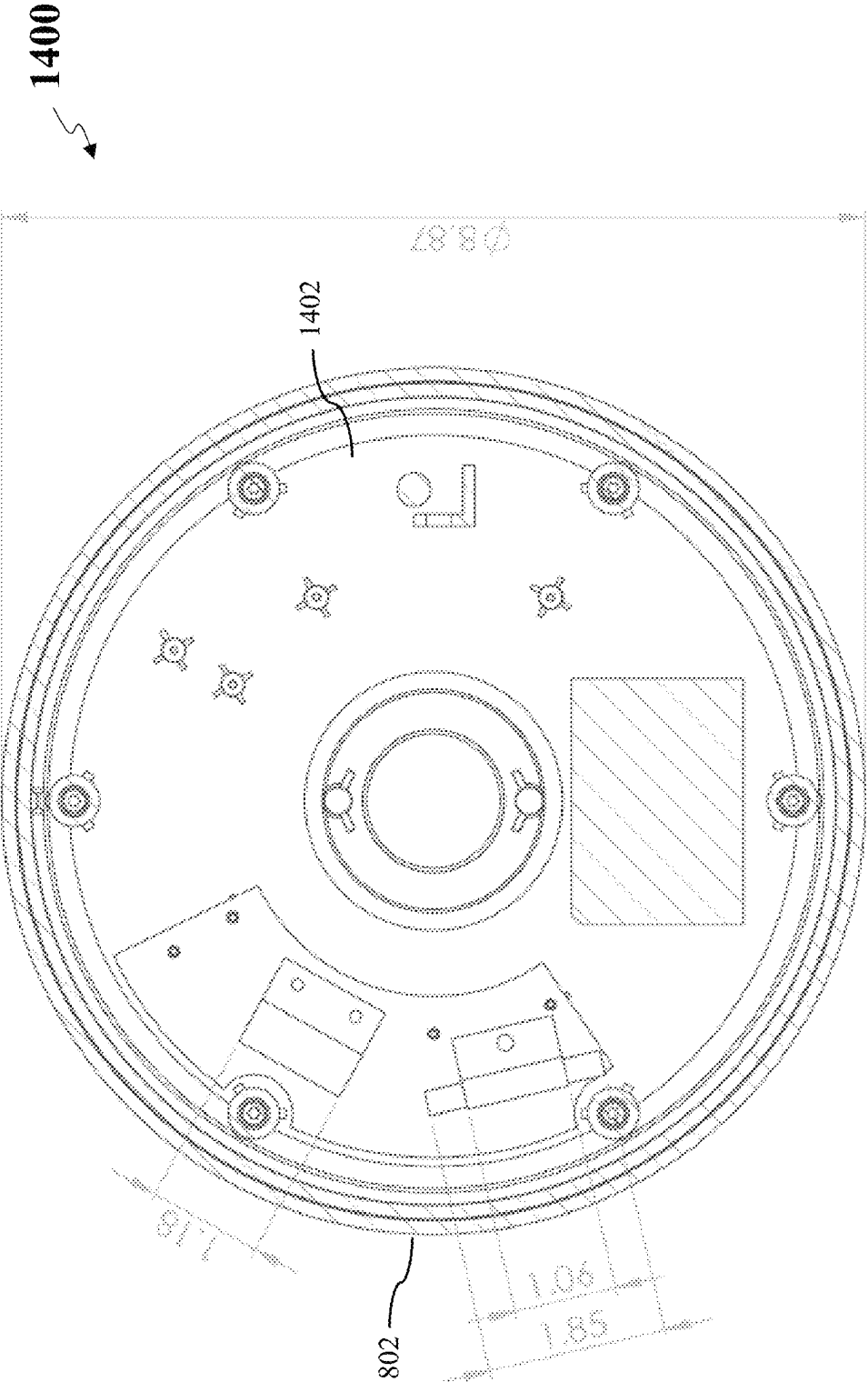


FIG. 14

1500

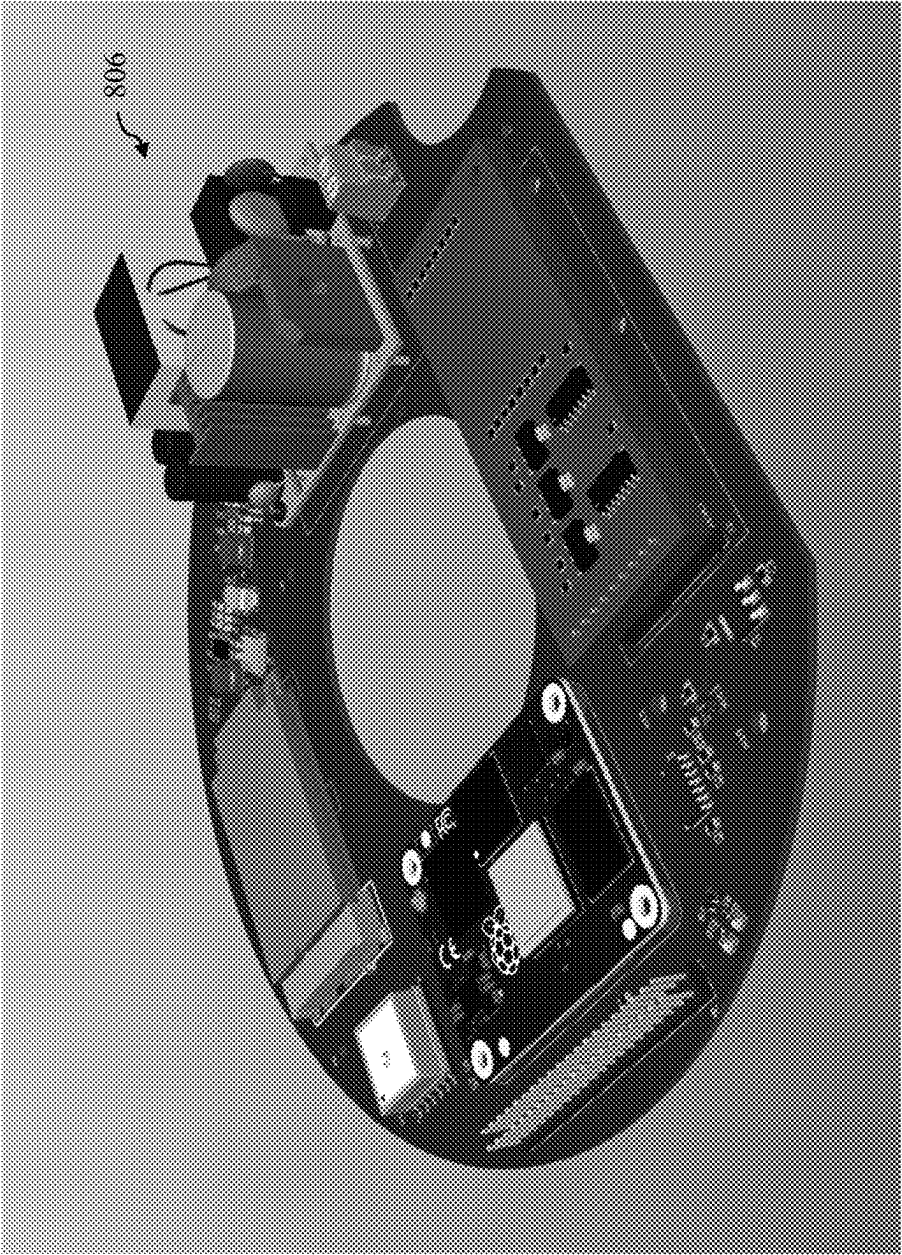


FIG. 15

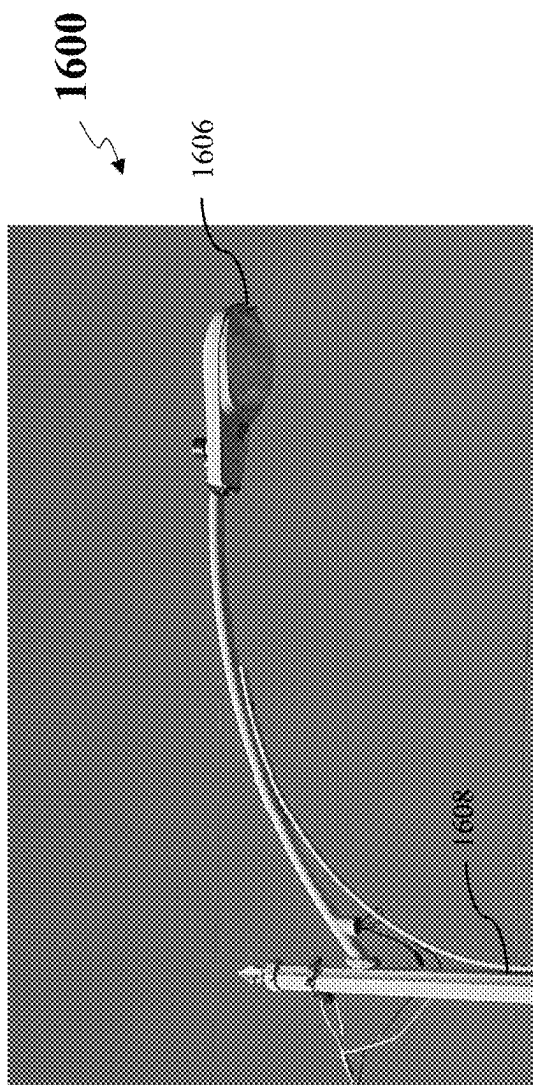


FIG. 16A

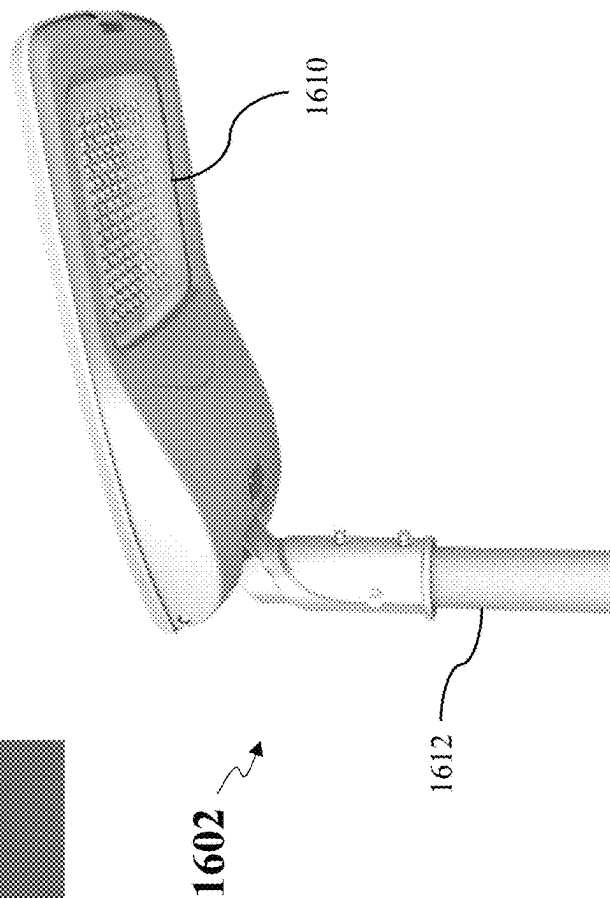


FIG. 16B

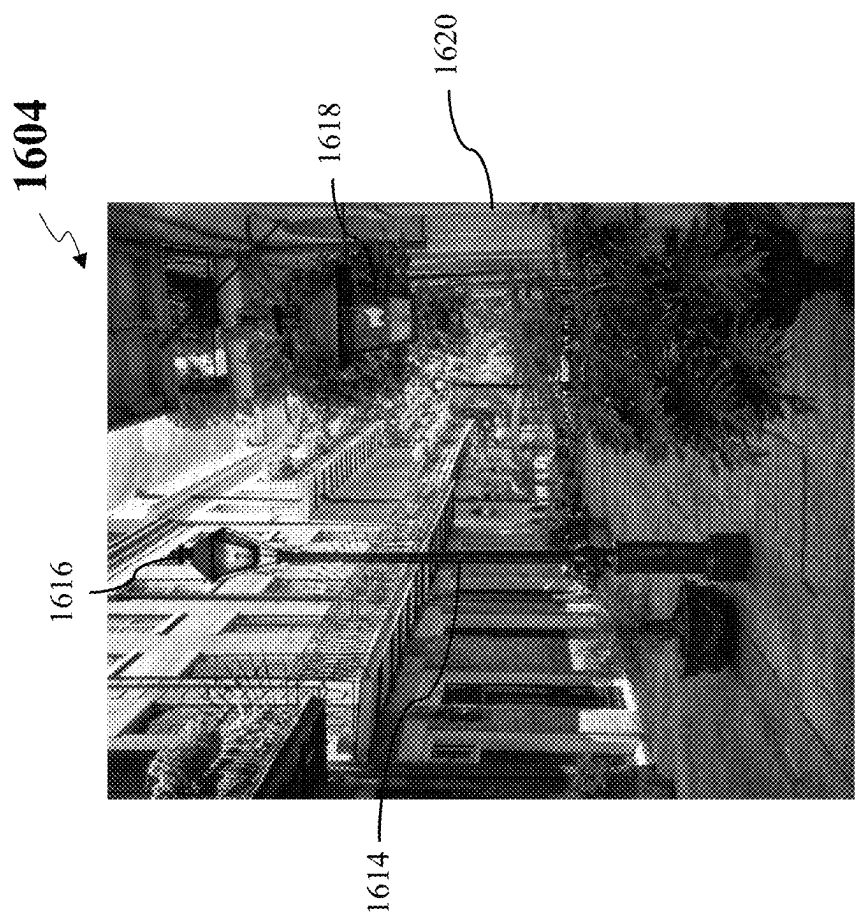


FIG. 16C

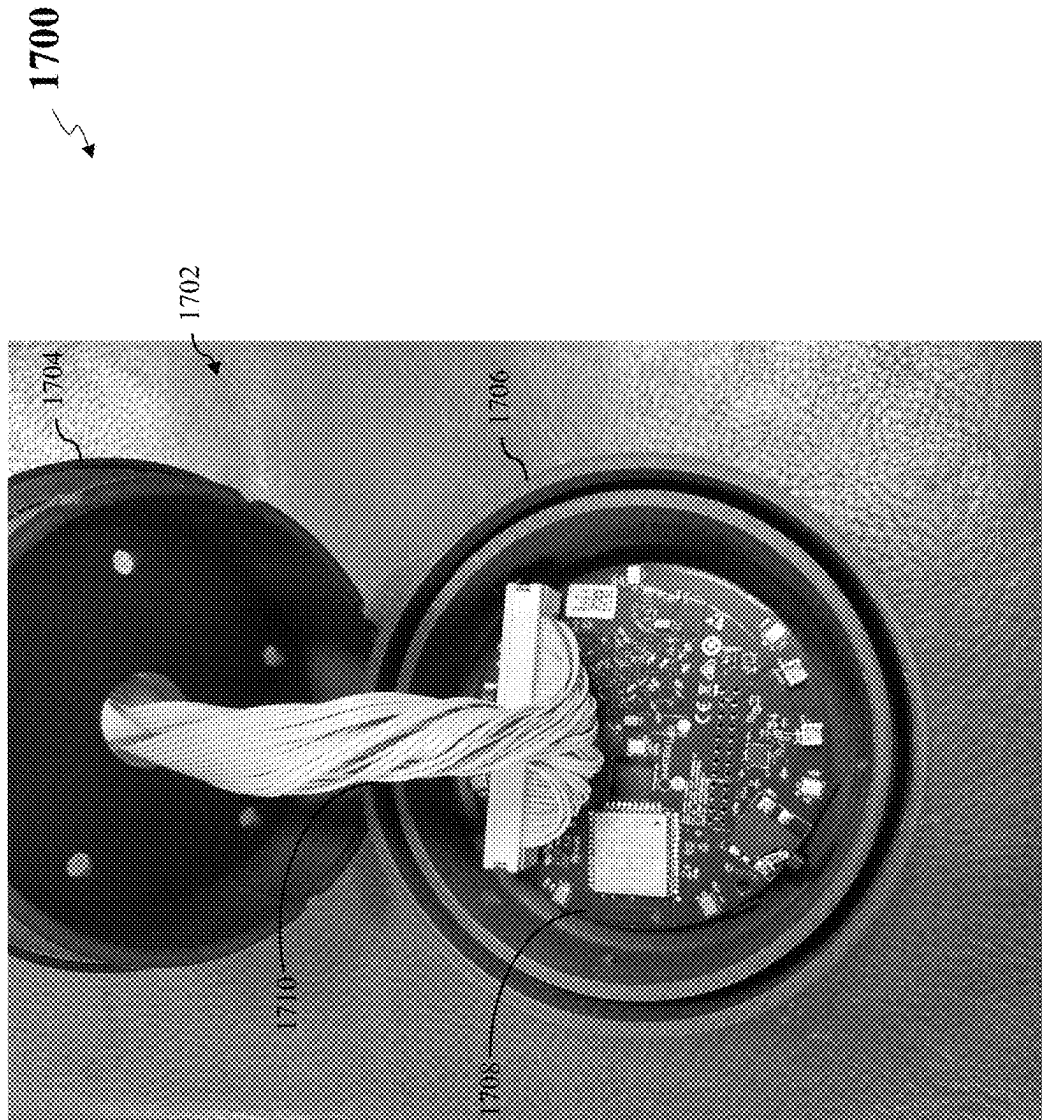


FIG. 17

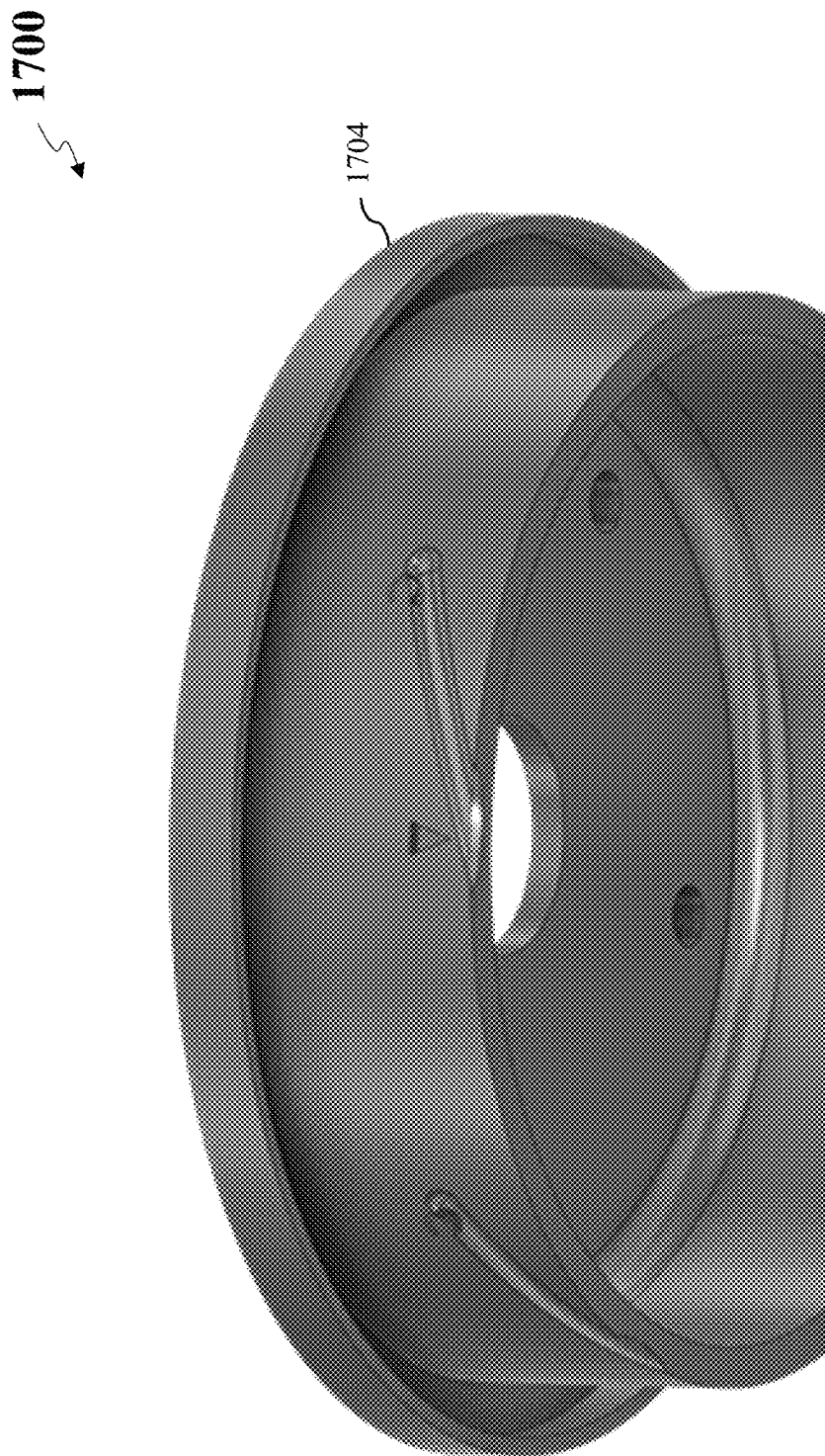


FIG. 18

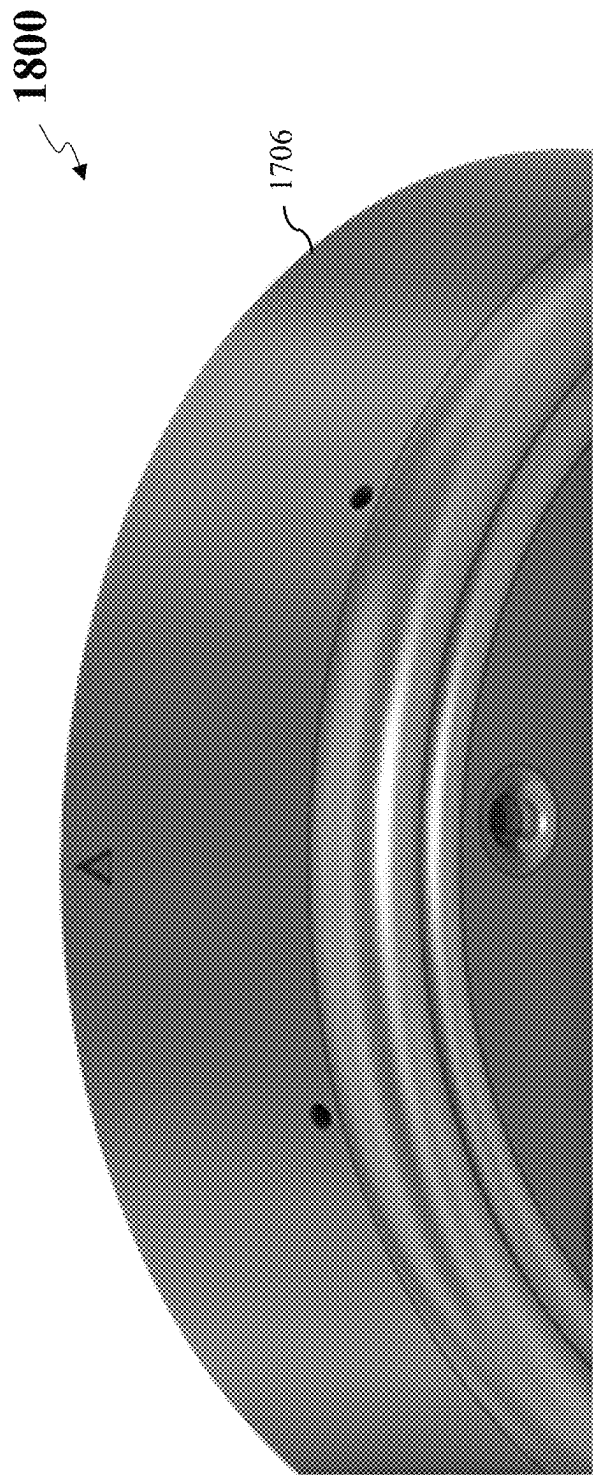


FIG. 19

2000
↘

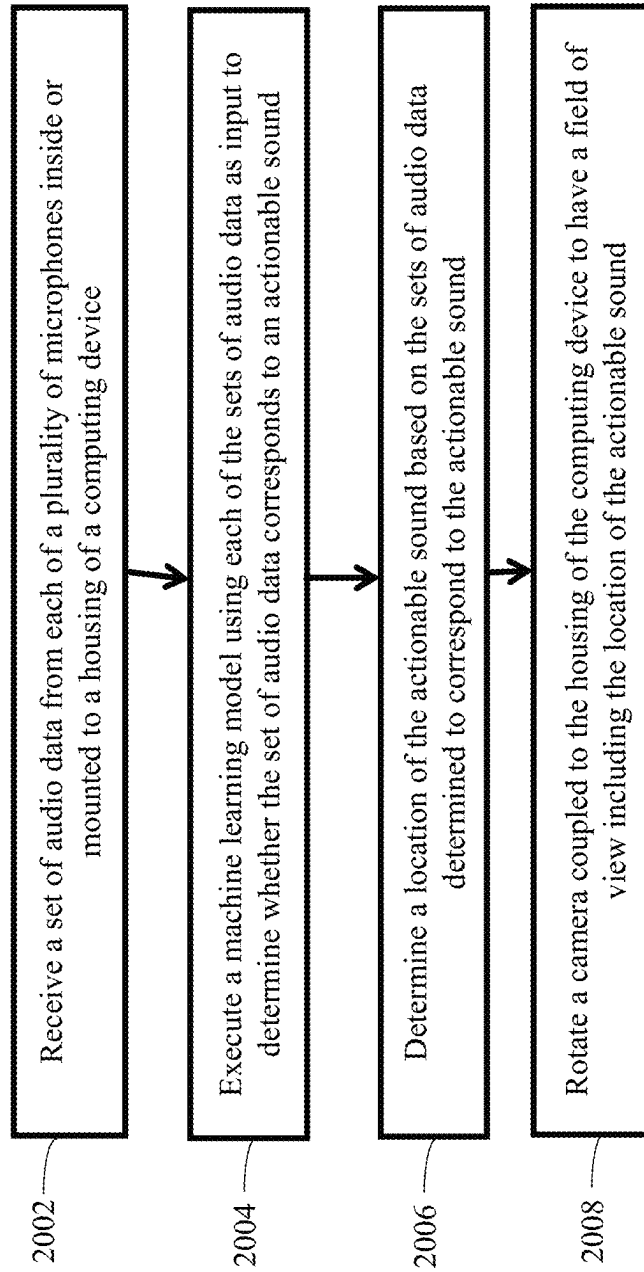


FIG. 20

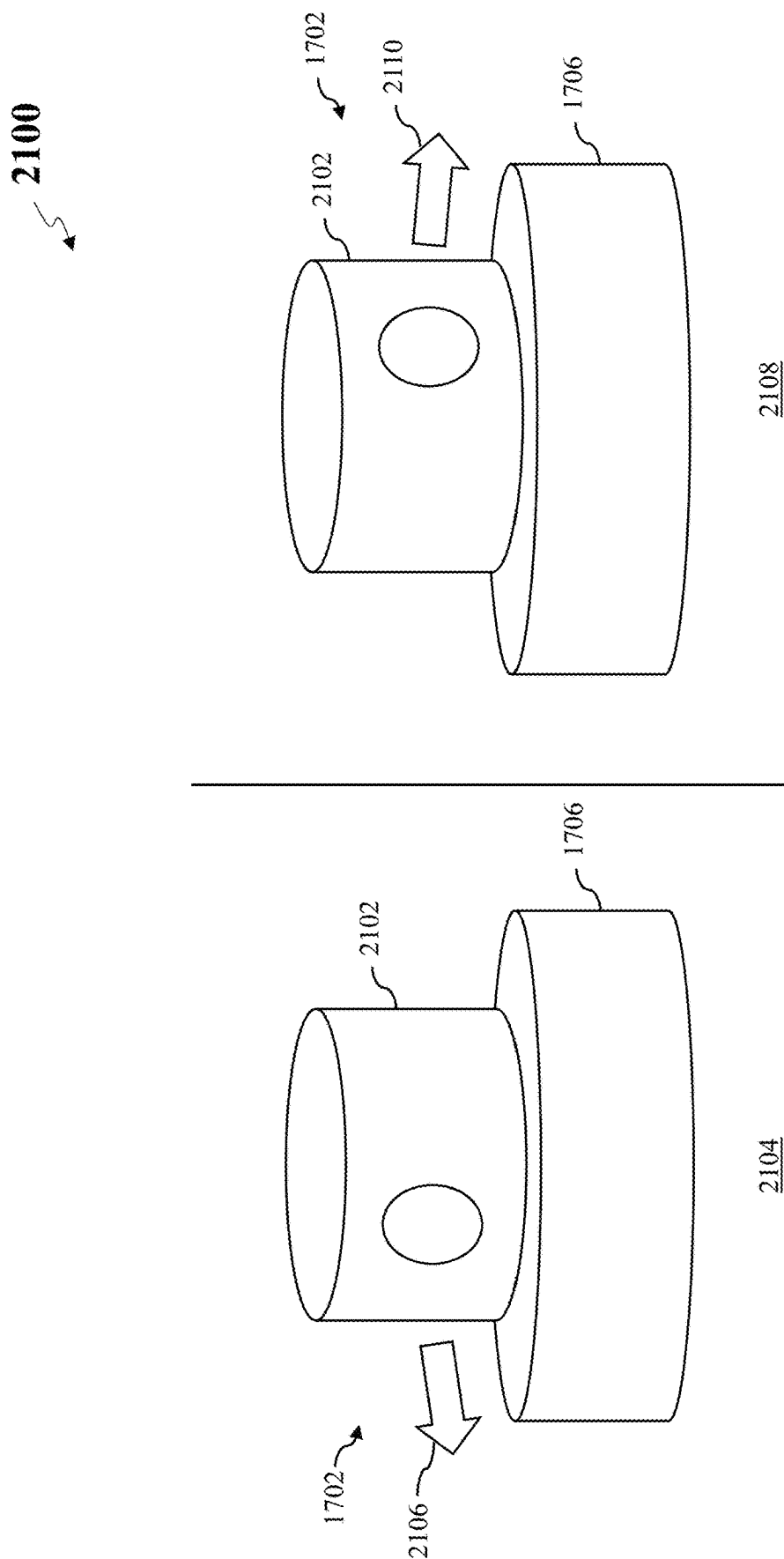




FIG. 22A

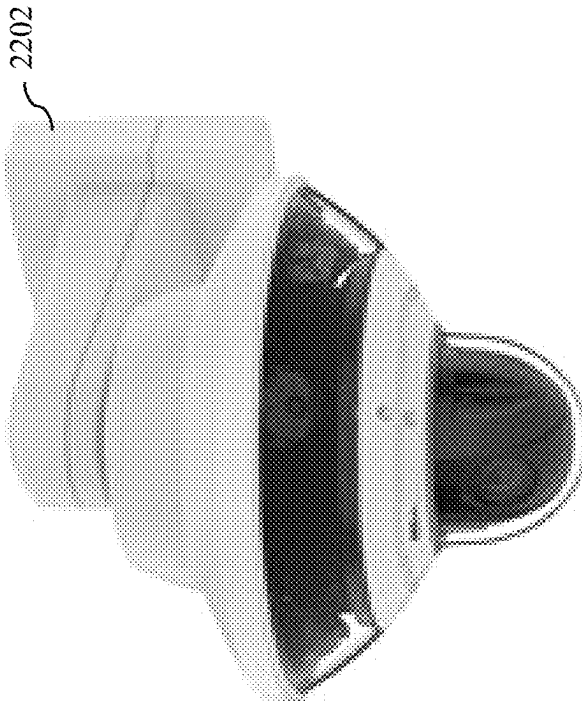


FIG. 22B

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SYSTEM AND METHOD FOR MULTILATERAL GUNSHOT DETECTION

TECHNICAL FIELD OF DISCLOSURE

The present invention relates generally to gunshot detection, and more particularly to gunshot detection using a mesh of recording devices.

BACKGROUND

Large cities or cities with large populations can host a significant amount of crime. The crime can be non-violent or violent crime and involve the use of firearms. During crimes involving firearms, a user may fire a firearm into the air or at various objects. Often times, authorities are notified about such a crime after the crime occurs and after the individuals involved have traveled away from the area. Even in instances in which authorities are notified of a crime while the crime is occurring, the notification may take time to travel to the authorities and the authorities may not know the location of the crime or the authorities may receive the incorrect location of the crime. These causes for delay may result in the individuals committing the crime not being identified and/or the individuals having time to remove any evidence of being at the scene.

SUMMARY

One solution to quickly identifying crimes involving firearms and the locations of such crime involves placing recording devices around a city. The recording devices may continuously stream audio data generated by the recording devices from sound of the surrounding areas to a remote server. The remote server can analyze the audio data to determine whether the audio data includes the sounds of at least one gunshot. If streamed audio from a recording device includes sounds of at least one gunshot, the remote server can identify the location of the device and determine the location is where the gunshot occurred.

Detecting a gunshot in the above-described manner can have a few technical drawbacks. For example, streaming audio across a network can require a significant amount of bandwidth. In a large city, a large number of recording devices may be required to canvas the entire city. Continuously streaming audio to the server from each device can require a significant amount of bandwidth on the network. Further, because of the large number of recording devices that stream the audio, continuously analyzing each of the streams may require a significant amount of processing power at the remote computing device to accurately detect gunshots. Additionally, continuously recording audio and transmitting to a centralized location can create the potential of capturing sounds other than gunshots which may or may not be legally or ethically captured.

Another problem is that an identification of a gunshot in audio streamed from a recording device may not enable detection of an accurate location of a gunshot. The sound waves of gunshots can be loud and can echo off of buildings surrounding the streets in a city. Accordingly, gunshots may travel a large distance in a city and multiple recording devices may record audio of the gunshot, even when the recording devices are not the closest recording devices to the location of the actual gunshot. This problem can be compounded when the recording devices are placed sporadically around the city, particularly when some recording devices are placed in locations in which the echoes can be detected

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and in other locations where the recording devices are blocked from recording sound. It can be difficult to predict a location of a gunshot if one recording device that is nearest to the gunshot is blocked from recording the gunshot and a recording device further away from the gunshot detects an echo of the gunshot.

Implementations of the systems and methods described herein may overcome the aforementioned technical problems. To do so, for example, a set of recording devices may be configured with housings that can be distributed across a metropolitan environment (e.g., a city) such that the recording devices are not blocked from recording audio. The housings of the recording devices can be mounted to light fixtures (e.g., light fixtures at the top of light posts or attached to buildings) that are interspersed throughout the metropolitan environment. One or more microphones may be mounted within or on the housings mounted to the light fixtures at the top of the light posts such that the microphones can record sounds within a radius of the housing with little impedance from passerby objects that recording devices at lower heights may encounter. The recording devices may include processors that are configured to continuously receive and process audio recordings from the microphones to determine when the audio recordings include audio of a gunshot.

A processor of a recording device may process received audio recordings by generating spectrograms from samples of the audio recordings at set or predetermined intervals. The processor may use machine learning techniques on the spectrograms to detect whether the spectrograms include audio data of a gunshot. As described herein, machine learning techniques may include deep learning techniques. The deep learning techniques can provide adaptive noise suppression by creating specialized digital filtration for each sensor's installed location, which can provide a silent steady state. For example, as the processor receives audio data from the microphone, the processor may generate spectrograms of the audio data for overlapping and/or non-overlapping time periods. The processor may execute a machine learning model with the spectrograms as input to determine whether the spectrograms include audio data of at least one gunshot. If the processor identifies a spectrogram that includes audio data of at least one gunshot, the processor may transmit the spectrogram or the audio data that is depicted in the spectrogram to a remote server (e.g., a cloud server) for further processing. In this way, the processor may avoid continuously streaming audio data to the remote server for processing, instead only sending audio data and/or a spectrogram of the audio data to the remote server upon determining the spectrogram is associated with a gunshot, reducing the bandwidth requirements of locating gunshots at the remote server.

The remote server may receive spectrograms and/or audio data associated with a gunshot from multiple recording devices and determine the location of the gunshot. The remote server may do so using a combination of machine learning and multilateration techniques. For example, the remote server may execute a machine learning model on the received spectrograms and/or generate spectrograms from received audio data and execute the machine learning model based on the generated spectrograms. In doing so, the machine learning model may identify portions or impulses of the spectrograms that correspond to gunshots. The remote server may identify times of detected gunshots from each of the spectrograms and use multilateration techniques on the received times and stored locations of the recording devices that transmitted the spectrograms and/or audio data. The

remote server may identify the location of the gunshot based on the multilateration techniques. The remote server may then transmit a notification of a time of the gunshot (e.g., a time in which the gunshot occurred or was first recorded) and/or the identified location to a server or computing device accessed by authorities (e.g., police officers or detectives) to inform the authorities of the gunshot.

By using machine learning models and multilateration techniques in this way, the recording devices and the remote server may quickly and accurately determine a time and location of a gunshot. Accordingly, the remote server may transmit a notification of the gunshot to authorities more quickly than systems using conventional technologies. The fast notification may enable the authorities to arrive at the site of the gunshot in time to potentially apprehend any bad actors at the site and/or to collect evidence of any crimes that were committed.

In one embodiment, an apparatus for detecting gunshots is disclosed. The apparatus may include a device housing configured to removably couple to a light post or a light fixture; a microphone inside or mounted to the device housing; and a processor inside the device housing and electrically coupled to the microphone. The processor can be configured to receive audio data from the microphone; execute a machine learning model using the audio data as input to determine whether the audio data corresponds to a gunshot; and responsive to determining the audio data corresponds to a gunshot, transmit the audio data to a remote processor.

In another embodiment, a method for detecting gunshots is disclosed. The method may include receiving, by a recording processor of an edge recording device, a first set of audio data from a microphone inside or mounted to a housing of the edge recording device, the first set of audio data comprising a sound recording; executing, by the recording processor, a first machine learning model using the first set of audio data as input to determine the first set of audio data is associated with a gunshot; responsive to determining the first set of audio data is associated with a gunshot, transmitting, by the recording processor, the first set of audio data to a first remote processor; receiving, by the first remote processor, the first set of audio data as a set of audio data of a plurality of sets of audio data received from a plurality of edge recording devices, each of the plurality of sets of audio data transmitted to the first remote processor in response to a determination the set is associated with a gunshot; iteratively executing, by the first remote processor, a second machine learning model using each of the plurality of sets of audio data as input to determine a time of a gunshot for each set of audio data; executing, by the first remote processor, a multilateration model using the time of the gunshot for each set of audio data to determine a location of a first gunshot; and transmitting, by the first remote processor, an indication of the location to a second remote processor of a second remote computing device.

In another embodiment, a system for detecting gunshots is disclosed. The system may include a first remote processor of a first remote computing device remote from a set of edge recording devices, the first remote processor coupled to a first remote non-transitory memory of the first remote computing device, wherein the first remote processor is configured to receive a set of audio data from each of a subset of the set of edge recording devices, each set of audio data transmitted to the remote processor in response to a determination that a gunshot is associated with the set of audio data; execute a machine learning model using each set of audio data as input to determine a time of a gunshot for

each set of audio data; execute a multilateration model using the time of the gunshot for each set of audio data and a location of each of the subset of edge recording devices as input to determine a location of the gunshot; and transmit an indication of the location to a second remote processor of a second remote computing device.

The system may further include a recording processor of an edge device of the subset of edge devices. The recording processor may be in communication with an edge non-transitory memory of the edge device and a microphone mounted on or in a housing of the edge device. The recording processor may be configured to receive a first set of audio data from the microphone, the first set of audio data comprising a sound recording; execute a second machine learning model using the first set of audio data as input to determine the first set of audio data is associated with a gunshot; and responsive to determining the first set of audio data is associated with a gunshot, transmit the first set of audio data to the first remote processor as a set of audio data from the subset of edge devices.

In another embodiment, an apparatus is disclosed. The apparatus may include a device housing; a camera; a plurality of microphones inside or mounted to the device housing; and a processor inside the device housing and electrically coupled to the plurality of microphones. The processor can be configured to receive a set of audio data from each of the plurality of microphones; execute a machine learning model using each of the sets of audio data as input to determine whether the set of audio data corresponds to an actionable sound; determine a location of the actionable sound relative to a location of the camera based on a plurality of sets of audio data determined to correspond to the actionable sound; and rotate the camera towards the determined location of the actionable sound.

In another embodiment, a method is disclosed. The method may include receiving, by a processor of a computing device, a set of audio data from each of a plurality of microphones inside or mounted to a housing of the computing device; executing, by the processor, a machine learning model using each of the sets of audio data as input to determine whether the set of audio data corresponds to an actionable sound; determining, by the processor, a location of the actionable sound relative to a location of a camera based on a plurality of sets of audio data determined to correspond to the actionable sound; and rotating, by the processor, a camera (e.g., a camera coupled to a housing of the computing device) to have a field of view including the location of the actionable sound.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting embodiments of the present disclosure are described by way of example with reference to the accompanying figures which are schematic and are not intended to be drawn to scale. Unless indicated as representing the background art, the figures represent aspects of the disclosure.

FIG. 1 illustrates components of a gunshot detection system, according to an embodiment.

FIG. 2 is a flow diagram illustrating an exemplary method for detecting gunshots, according to an embodiment.

FIG. 3 is a sequence diagram of a sequence for gunshot detection, according to an embodiment.

FIG. 4 is another sequence diagram of a sequence for gunshot detection, according to an embodiment.

FIG. 5 is an illustration of a network mesh of a gunshot detection system, according to an embodiment.

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FIG. 6 is an illustration of locations of recording devices of a gunshot detection system within a metropolitan area, according to an embodiment.

FIG. 7 illustrates depictions of a set of audio data, according to an embodiment.

FIG. 8 illustrates a housing for a recording device, according to an embodiment.

FIG. 9 illustrates a perspective view of a top enclosure of the housing of FIG. 8, according to an embodiment.

FIG. 10 illustrates a side view of the housing of FIG. 8, according to an embodiment.

FIG. 11 illustrates a cross-sectional view of the housing of FIG. 8, according to an embodiment.

FIG. 12 illustrates a top view of a printed circuit board of FIG. 8, according to an embodiment.

FIG. 13 illustrates a top view of a top surface of a bottom enclosure of FIG. 8, according to an embodiment.

FIG. 14 illustrates a bottom view of a bottom surface of a top enclosure of FIG. 8, according to an embodiment.

FIG. 15 illustrates a perspective view of a printed circuit board of FIG. 8, according to an embodiment.

FIGS. 16A-C illustrate perspective views of different light fixtures coupled to light posts to which a recording device can be coupled, according to an embodiment.

FIG. 17 illustrates a perspective view of a recording device, according to an embodiment.

FIG. 18 illustrates a perspective view of a top enclosure of a housing of the recording device of FIG. 17, according to an embodiment.

FIG. 19 illustrates a perspective view of a bottom enclosure of a housing of the recording device of FIG. 17, according to an embodiment.

FIG. 20 is a flow diagram illustrating an exemplary method for detecting actionable sounds, according to an embodiment.

FIG. 21 illustrates a sequence of rotating a camera of a recording device in response to detecting a gunshot, according to an embodiment.

FIGS. 22A-B illustrate different camera types that can be rotated in response to detecting a gunshot, according to an embodiment.

DETAILED DESCRIPTION

The present disclosure is here described in detail with reference to embodiments illustrated in the drawings, which form a part hereof. Other embodiments may be used and/or other changes may be made without departing from the spirit or scope of the present disclosure. The illustrative embodiments described in the detailed description are not meant to be limiting of the subject matter presented here.

Reference will now be made to the exemplary embodiments illustrated in the drawings, and specific language will be used here to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Alterations and further modifications of the inventive features illustrated here, and additional applications of the principles of the inventions as illustrated here, which would occur to one skilled in the relevant art and having possession of this disclosure, are to be considered within the scope of the invention.

A computing system involving recording devices interspersed throughout a metropolitan area and a cloud server can operate together to detect and locate gunshots within the metropolitan area. In a non-limiting example, microphones of recording devices mounted to light fixtures (e.g., light fixtures on light posts or sides of buildings) throughout a city

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can record audio data of a gunshot. Such light posts can be utility poles. The recording devices may each execute a machine learning model to determine the audio data includes sounds of a gunshot and transmit the audio data to a cloud server. The cloud server may receive and analyze the audio data from the different recording devices to determine the times in which each recording device recorded the audio data (e.g., the times of arrival). Based on the times, the cloud server can determine the location and/or time in which the gunshot occurred. The cloud server can generate a notification containing the location and/or time and transmit the notification to a computing device operated by authorities, thus informing the authorities of the gunshot. FIG. 1 depicts an example environment that includes example components of a system that includes such recording devices and such a cloud server. Various other system architectures may include more or fewer features and/or may utilize the techniques described herein to achieve the results and outputs described herein. Therefore, the system depicted in FIG. 1 is a non-limiting example.

FIG. 1 illustrates a gunshot detection system 100, according to an embodiment. The system 100 may include an analytics server 110a, a system database 110b, recording devices 120a-c (collectively, recording devices 120 and individually, recording device 120), an administrator computing device 130, and/or a controller computing device 140. The above-mentioned components may be connected to each other through a network 150. Examples of the network 150 may include, but are not limited to, private or public LAN, WLAN, MAN, WAN, and the Internet. The network 150 may include wired and/or wireless communications according to one or more standards and/or via one or more transport mediums.

The communication over the network 150 may be performed in accordance with various communication protocols such as Transmission Control Protocol and Internet Protocol (TCP/IP), User Datagram Protocol (UDP), and IEEE communication protocols. In one example, the network 150 may include wireless communications according to Bluetooth specification sets or another standard or proprietary wireless communication protocol such as 802.11ah. In another example, the network 150 may also include communications over a cellular network, including, e.g., a GSM (Global System for Mobile Communications), CDMA (Code Division Multiple Access), and/or EDGE (Enhanced Data for Global Evolution) network.

The gunshot detection system 100 is not confined to the components described herein and may include additional or other components, not shown for brevity, which are to be considered within the scope of the embodiments described herein.

The analytics server 110a may be any computing device comprising a processor and non-transitory machine-readable storage capable of executing the various tasks and processes described herein. Non-limiting examples of such computing devices may include workstation computers, laptop computers, server computers, and the like. While the system 100 includes a single analytics server 110a, the analytics server 110a may include any number of computing devices operating in a distributed computing environment, such as a cloud environment.

The system database 110b may be a relational database or any other type of database. The system database 110b may be stored in memory of the analytics server 110a or one or more other computing devices. The system database 110b may store data or information about recording devices 120. For instance, the system database 110b may store the loca-

tions of the recording devices **120**, identifications or identifiers (e.g., numerical or alphanumeric identifications or identifiers) of the recording devices **120**, and/or identifications or identifiers of connections the analytics server **110a** has with the recording devices **120**.

The recording devices **120** may be computing devices similar to the analytics server **110a** that are located at various locations around a metropolitan area. The recording devices **120** may include processors, memory, and/or microphones that are attached to or mounted in housings (e.g., the housings shown and described with reference to FIG. **8**). The microphones may be or include one or more omnidirectional microphones configured to detect sounds from multiple directions (e.g., from all directions). The housings may be removably coupled to light fixtures (e.g., municipal street light fixtures) on light posts **125a-c** (collectively, light posts **125** and individually, light post **125**) around the metropolitan area. For example, the housings may fasten (e.g., through a “twist lock” plug and socket system) to light fixtures at the tops of the light posts **125**. By fastening to the light fixtures at the tops of the light posts **125**, the microphones attached to the housings may better capture the sounds of the surrounding environment because the microphones are higher and not blocked by objects or pedestrians that walk past the microphones. Further, because the housings are on top of light posts **125** instead of at ground level, the recording devices **120** may be less accessible to malicious individuals that may attempt to damage or tamper with the recording devices **120**.

The recording devices **120** may be powered by the same or a common power source to the light posts **125** to which the recording devices **120** are coupled. For example, a light post **125** may be electrically connected to a power grid that provides power to the metropolitan area. A recording device **120** connected to the light post **125** may couple directly to the light post **125** in series or parallel to receive power from the grid. In some cases, the recording device **120** may be directly connected to the power grid through a power line. Each of the recording devices **120** around the metropolitan area may be similarly coupled to the power grid. In this way, the recording devices **120** may receive adequate power to power the processors and microphones of the recording devices **120** for continuous recording and/or processing of audio data.

The recording devices **120** may not store the audio recordings that the recording devices **120** generates. Instead, the recording devices **120** may generate the recordings, analyze the recordings to determine if the recordings include audio of any gunshots, and discard any recordings that do not contain gunshot audio. In some cases, the recording devices **120** may transmit audio segments (or spectrograms of such audio segments) the recording devices **120** determine contain audio data of gunshot to the analytics server **110a**. The recording devices **120** may discard such audio segments from memory subsequent to transmitting the audio segments to the analytics server **110a**. Accordingly, the recording devices **120** may conserve memory resources.

The housings of the recording devices **120** may be configured to receive photoelectric cells. The photoelectric cells may operate to detect light over the course of a day and night. The photoelectric cells may operate for security to cause lights (e.g., light bulbs) of the light posts **125** to turn on at night when the sun is not out and off during the day when the sun is out. The photoelectric cells may be powered by the power grid or another power source common to the recording devices **120** and/or the lights of the light posts **125**. In some cases, a photoelectric cell, the recording device

120, and the light of the light post **125** can be powered by separate power sources (e.g., the recording device **120** and the photoelectric cell coupled to the recording device **120** may be powered by a stored battery within the recording device **120** while the light may be powered by the energy grid).

The microphones of the recording devices **120** may continuously record audio data (e.g., sounds) of their surrounding environments. The microphones may record the audio data over time. As the microphones are recording the audio data, the microphones may transmit the audio data to the processors of recording devices **120** (e.g., the processors mounted to or inside the same housings as the microphones).

The processors of the recording devices **120** may receive and process the audio data from the recording devices **120**. For example, a processor of a recording device **120** may receive audio data that a microphone of the recording device **120** recorded over the course of a time frame. The processor may process the audio data by generating a spectrogram from the audio data that illustrates the sound wave and/or frequencies of the audio data from the time frame.

The spectrogram may be a visual diagram of the strength of the signal strength (e.g., loudness) of a sound signal (e.g., a sound wave) over time at various frequencies. The spectrogram may be a visual diagram of the strength of the signal strength (e.g., loudness) of a sound signal (e.g., a sound wave) over time at various frequencies. The spectrogram or multiple spectrograms from the same event (e.g., same gunshot) may be also analyzed for total envelope content, timing, as well as evidence of specific characteristics that identify pertinent information regarding the weapon(s), round(s) or shooter(s) involved in the event. A spectrogram or multiples thereof may also be archived and forensically analyzed for comparison to other events to identify similarities between events. A machine learning model (e.g., a deep learning model) may provide scoring on the similarity between events.

The processor may execute a machine learning model (e.g., a neural network, a support vector machine, a random forest, etc.) that has been trained to analyze spectrograms to detect gunshots. The processor may execute the machine learning model with the spectrogram to determine whether the spectrogram (e.g., the spectrogram as a whole) illustrates audio data of a gunshot (e.g., includes sounds of at least one gunshot). The machine learning model may output a binary output (e.g., a two-class output) indicating whether the audio data includes audio of a gunshot or not.

In some cases, the machine learning model may output confidence scores indicating a likelihood that the spectrogram includes audio data of a gunshot. The machine learning model may output a confidence score indicating the likelihood that the spectrogram includes audio data of a gunshot and/or a confidence score indicating the likelihood that the spectrogram does not include audio data of a gunshot. The machine learning model or the processor executing the machine learning model may compare the confidence score for the likelihood that the spectrogram includes audio data of a gunshot to a threshold (e.g., a defined threshold). If the confidence score is higher than the threshold, the machine learning model or the processor may determine the spectrogram likely includes audio data of a gunshot. Otherwise, the machine learning model or the processor may determine the spectrogram likely does not include audio data of a gunshot.

Upon determining the audio data likely or does include audio data of the gunshot, the recording device **120**, may transmit the audio data and/or the spectrogram of the audio data to the analytics server **110a**. The analytics server **110a**

may receive the audio data and/or the spectrogram from the recording device **120** and/or other recording devices **120** that similarly recorded and detected audio data that contains a gunshot. The analytics server **110a** may generate spectrograms from any audio data (e.g., sets of audio data received from individual recording devices **120**) the analytics server **110a** receives from the recording devices **120**. The analytics server **110a** may process the audio data and/or spectrograms to determine a time and/or a location of the gunshot. For instance, the analytics server **110a** may store a machine learning model (e.g., a neural network, support vector machine, random forest, etc.) that is trained to use object recognition techniques to detect gunshots at individual pulses or times of spectrograms. The analytics server **110a** may iteratively execute the machine learning model with each of the spectrograms the analytics server **110a** receives from the recording devices **120**. The machine learning model may output times of one or more gunshots from each spectrogram upon being executed. The times may indicate the times (e.g., the times of arrival of the sounds of the gunshots) in which each recording device **120** recorded gunshots within the time periods of the spectrograms.

The analytics server **110a** may identify the times of the recorded gunshots and use the times to determine the locations of the gunshots. The analytics server **110a** may do so using the identified times and stored locations of the recording devices **120** that recorded the gunshots. For instance, the analytics server **110a** may identify the recording devices **120** that transmitted the spectrograms or the audio data of the gunshots. The analytics server **110a** may do so by identifying identifiers of the connections through which the analytics server **110a** received the audio data and/or spectrograms or by identifying identifiers (e.g., IP addresses) in the data packets that contained the received audio data and/or spectrograms. The analytics server **110a** may use the identifiers in a look-up technique through the system database **110b** to identify the locations of the recording devices **120** that transmitted the spectrograms and/or audio data of the recorded gunshots.

The analytics server **110a** may execute a multilateration model to determine a location of a gunshot based on the detected times of the gunshots and the locations of the recording devices **120** that detected the gunshots. The multilateration model may be or include executable instructions stored by the analytics server **110a** in memory that, upon execution, determines the locations of gunshots. For example, executing the multilateration model may cause the multilateration model to calculate a potential location of the gunshot for individual groupings of three recording devices **120** (e.g., groupings of recording devices that transmitted audio data of a gunshot). The multilateration model may do so, for instance, by applying the following set of equations:

$$(x_s - x_1)^2 + (y_s - y_1)^2 = (r_s - s_1 r_1)^2$$

$$(x_s - x_2)^2 + (y_s - y_2)^2 = (r_s - s_2 r_2)^2$$

$$(x_s - x_3)^2 + (y_s - y_3)^2 = (r_s - s_3 r_3)^2$$

where r is the time difference of arrival, x and y are the coordinates (e.g., the geographic locations), and s is the solution for the variable. The detected times of the gunshots and the locations of the recording devices **120** in the groupings of three recording devices **120** may be used as input into the equations input. In this way, the analytics server **110a**, may calculate potential locations of the gunshot detected by the different recording devices **120**.

Upon calculating the potential locations, the multilateration model may iteratively filter out the different groupings of recording devices **120**. For example, the multilateration model may calculate an average gunshot location from the calculated potential gunshot locations of groupings of three recording devices **120** that detected the gunshot. The multilateration model may then use a distance formula between the potential locations of the gunshot for each grouping of recording devices **120** to calculate distances between the potential locations of the groupings and the average location of all of the groupings. The multilateration model may identify the grouping or a set number of groupings associated with the largest distance or distances and remove the identified grouping or set from the list. In some cases, the multilateration model may remove any groupings from the list that are associated with a distance that exceeds a threshold. The multilateration model may then calculate a new average location from the potential locations of the remaining recording devices **120** on the list. The multilateration model may calculate distances between the potential locations and the new average location for the remaining groupings of recording devices **120** and remove any groupings of recording devices **120** that are associated with a largest calculated distance or a distance that exceeds the same or a different threshold. The multilateration model may repeatedly perform this process until only one grouping of recording devices **120** is left on the list. The multilateration model may identify the potential location of the remaining grouping as the location of the gunshot.

The analytics server **110a** may identify and transmit the location of the shot to the administrator computing device **130**. In doing so, the analytics server **110a** may transmit the location and/or a timestamp indicating a time of the gunshot to the administrator computing device **130**. The administrator computing device **130** may be any computing device that is owned and/or accessed by authorities (e.g., police officers or detectives). The authorities may view the location and/or the time stamp at a user interface and travel to the location (e.g., travel to the location to investigate what caused the gunshot to occur and/or the outcome of the gunshot).

In some cases, the analytics server **110a** may transmit the location and/or the timestamp of the gunshot to the controller computing device **140**. The controller computing device **140** may be any computing device that is configured to control a device **145** (e.g., a drone (an unmanned vehicle) such as an aerial drone or grounded drone equipped with a camera to capture images and/or videos of different areas). The analytics server **110a** may transmit a message including the location and/or the timestamp to the controller computing device **140**. The controller computing device **140** may, in some cases automatically upon receipt of the message, transmit instructions including the location and/or directions to the location, to the device **145** to cause the device **145** to travel to the location. The device **145** may capture images and/or video of the location upon reaching the location. In doing so, the device **145** may capture evidence of the gunshot and any crimes that were committed with the shooting of the gunshot by capturing images of the individuals involved and/or the surroundings of the gunshot (e.g., broken glass that has not been swept up yet, images of individuals scrubbing fingerprints, etc.). Because the process may occur automatically and in real-time, the device **145** may be able to capture evidence of a crime relatively quickly compared with conventional systems and/or methods.

The decrease in processing time to identify the location of the shooting event may aid in dispatching high speed evidence collecting resources (Drone as a First Responder

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“DFR”). For example, in some cases, the device **145** may capture infrared/thermal imagery of the scene prior to items cooling down, identifying within its field of view the location of high temperature elements in the vicinity of the crime scene (e.g., bullet casings, recently fired weapons, living or recently living bodies, running vehicles, recently running vehicles, etc.). Times and temperatures are recorded by a camera of the device **145** for each element of interest. This imagery can be overlaid with visible light imagery to provide investigators with actionable data relevant to the crime scene. Often dramatically reducing the time required to process the scene and respond to the crime. The imagery can also be provided live to first responders to determine if the event is still underway or over, and whether to dispatch SWAT, ambulances or investigators.

In some cases, one or more of the recording devices **120** may include cameras that are controlled by the processors of the recording devices **120**. The cameras may continuously capture images or video of the areas surrounding the recording devices **120**. The processors of the recording devices **120** may control the cameras (e.g., rotate or change the states of the cameras from on to off or vice versa). The processors may do so based on audio data the processors receive from the microphones of the recording devices **120**. For example, a processor of a recording device **120** may determine a set of audio data contains a sound of a gunshot (or another quick and loud (e.g., sharp) noises such as car accidents, explosions, screams, etc., using a machine learning model). Upon determining the set of audio data contains a sound of a gunshot, in addition to or instead of transmitting the audio data to the analytics server **110a**, the processor may determine a location of the gunshot. Upon determining the location, the processor may generate and/or transmit a control signal (e.g., a control signal containing identifications of elevation and azimuth) to the camera to control and/or rotate the camera in the direction or to the location of the gunshot. Accordingly, the processor can cause the camera to capture images of the aftermath of the gunshot, which may be useful to authorities for gathering evidence of a crime.

To determine the location of the gunshot, the recording devices **120** may include multiple sensors that are mounted on the housings of the recording devices **120**. The processors of the recording devices **120** may use multilateration techniques based on times of arrival of the gunshot sound at each of the multiple sensors on the housings of the recording devices **120**. For example, a recording device **120** may include five sensors that are mounted in a housing. Each sensor may detect a sound of a gunshot. The sensors may transmit the gunshot sound in audio data to a processor of the recording device **120**. The processor may generate a spectrogram from the audio data for each sensor. The processor may execute a machine learning model with the spectrograms to determine if the audio data contains audio data of a gunshot. Upon determining the spectrograms contain audio data of a gunshot, the processor may execute another machine learning model to detect times of the gunshots from the spectrograms (e.g., times of impulses of gunshot audio from the spectrograms). The processor may then use the multilateration techniques described herein based on the times of the gunshots in each of the spectrograms and the locations of the microphones within the housing of the recording device. The processor may determine the times of the gunshots and use multilateration techniques to determine a location of the gunshot using the same or similar techniques to the analytics server **110a**.

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Upon determining the location of the gunshot, the processor of the recording device **120** may determine the direction of the gunshot relative to the determined location. To do so, for example, the processor may compare the location of the recording device **120** (e.g., the geographical location of the recording device **120**) with the determined location of the gunshot. The processor may determine a vector from the location of the recording device **120** to the location of the gunshot. The processor may transmit the vector to the camera to cause the camera to rotate to point to the location. In some cases, the processor may compare the vector to a vector indicating the current direction in which the camera is pointing. In such cases, the processor may determine a difference between the two vectors to generate a change in position vector or rotation vector. The processor may transmit the change in position vector or rotation vector to the camera to cause the camera to rotate the amount of the change in position vector or rotation vector. In this way, the processor of the recording device **120** may control the camera to point at locations of gunshots upon detecting the gunshots.

In instances in which the camera is in an off state (e.g., not capturing images or recording), a processor of a recording device **120** may additionally change the state of the camera to an “on” state (e.g., capturing images or recording). For example, upon determining a location of a gunshot, the processor may transmit a control signal to the camera to rotate the camera to capture images and/or video of the location of the gunshot. The processor may additionally transmit a control signal to the camera that causes the camera to turn on and/or begin capturing images and/or video. The control signal may cause the camera to remain on and/or capture images and/or video indefinitely until the camera is manually turned off or to do so for a defined amount of time. The camera may capture the images and/or video and transmit the images and/or video to the processor. The processor may store the images and/or video in memory for later retrieval. In this way, the processor may save memory storage and/or energy requirements of operating a camera by only capturing images and/or video when the images and/or video may be relevant to a gunshot investigation.

In some cases, upon determining the direction or location of the gunshot or that a gunshot occurred, the recording device **120** may transmit a message to the administrator computing device **130**. The message may indicate that a gunshot was detected at the recording device **120** and/or the determined location of the gunshot. In some cases, the recording device **120** may transmit images or video the camera captured of the determined location of the gunshot. In some cases, the video may be a livestream of the area. An operator at the administrator computing device **130** may view the images, video, and/or livestream and determine whether to take any action regarding the gunshots. In some cases, the recording device **120** may transmit an indication of the location to the controller computing device **140**. Upon receipt of the indication, the controller computing device **140** may control the device **145** to move the device **145** to the location to capture images or video of the location. Such imagery can be provided live to first responders to determine if an event (e.g., a gunshot, accident, fight) is still underway or over, and whether to dispatch SWAT, ambulances, or investigators.

In some cases, the recording device **120** may store and execute a machine learning model that is configured to detect auto-accidents. In such cases, the recording device **120** may record audio data and insert the audio data into the

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machine learning model as described herein (e.g., insert spectrograms of the audio data into the machine learning model). The recording device **120** may execute the machine learning model based on the audio data and the machine learning model may output an indication of whether the audio data contains audio of an auto-accident. Responsive to detecting an indication that the audio data contains audio of an auto-accident, the recording device **120** may determine a location auto-accident using the systems and methods described herein and control the camera to capture imagery of the location of the auto-accident. The recording device **120** may livestream the imagery to a computer accessed by first responders such that the first-responders can view of the scale and scope of the auto-accident. In some cases, the recording device **120** may transmit an indication of the location of the auto-accident to the controller computing device **140** to control the device **145** to move the device **145** to the location of the auto-accident. The recording device **120** may be similarly configured to identify and capture imagery or transmit locations of calls for help or duress. Because the recording device **120** can begin recording upon detection of an event (e.g., an event that produced an actionable sound, such as a gunshot, a car crash, or a scream), the recording device **120** can capture video of the event prior to an operator paying attention to the livestream. Accordingly, the operator may later playback and/or review the recorded video to have more information about the cause of the event.

The recording device **120** can control nearby cameras to rotate to the source of an event (e.g., a gunshot, an auto-accident, a scream, etc.). For example, the recording device **120** can determine the location of the source of the sound relative to the recording device **120** and/or relative to the camera. The recording device **120** can transmit a control signal including an azimuth and an elevation to the camera (or a device controlling the camera) to cause the camera to rotate towards the location of the source of the location, thus managing the orientation of the camera. The control signal may trigger the camera to begin recording and/or livestreaming to the video management service. The recording device **120** can then transmit a message to a video management service (e.g., a computer accessed by the authorities) indicating the camera has been rotated and/or that event was detected within range of the camera, thus alerting any individuals accessing the computer that receives the message of the rotation and to begin viewing the livestream provided by the camera. Alternatively, the recording device **120** can transmit a message to the third party video management system that includes the location of the source of the event. The recording device **120** may include an elevation and/or azimuth (and/or pan, tilt, or zoom instructions) for a camera to rotate to capture the source of the event. The third-party video management system can receive the location and/or elevation and/or azimuth and control a camera or cameras (e.g., the camera or cameras closest and/or in a defined radius of the source of the event) to capture the source of the event (e.g., begin live streaming the source of the event). In this way, the recording device **120** can rotate or manage the orientation of the camera and/or trigger the camera to begin recording. In some cases, the recording device **120** can control the camera or transmit a message of a location and/or azimuth and/or elevation to capture a lane of egress associated with the event.

FIG. 2 is a flow diagram illustrating an exemplary method **200** for detecting gunshots, according to an embodiment. The method **200** may include steps **202-214**. However, other embodiments may include additional or alternative steps, or

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may omit one or more steps altogether. The method **200** is described as being executed by a data processing system (e.g., the analytics server **110a**, as described with reference to FIG. 1) and/or a recording device (e.g., the recording device **120**). However, one or more steps of the method **200** may be executed by any number of computing devices operating in the distributed computing system. For instance, one or more computing devices may locally perform part or all of the steps described in FIG. 2 or a cloud device may perform such steps.

In step **202**, the recording device may receive a first set of audio data from a microphone. The microphone may be inside or mounted to a housing of the recording device. The first set of audio data may be or include a sound recording from audio collected within a particular time frame. The microphone may continuously record sounds made in the environment surrounding the recording device. As the microphone is recording the sounds, the microphone can transmit the recorded sounds to a processor of the recording device for further processing.

The recording device may receive the first set of audio data and/or generate a spectrogram from the first set of audio data. The recording device may generate the spectrogram for the first set of audio data, for example, by calculating the Fourier transforms of various segments (e.g., equal-length segments) of the audio data in which the frequency of the audio data does not substantially change. The recording device may then generate a graph of the output of the Fourier transform calculation. The graph may illustrate the decibels of the audio data at different frequencies.

When generating the spectrogram for the first set of audio data, the recording device may include timestamps in the spectrogram of the real-world times in which the respective audio data was generated or received from the microphone or plurality of microphones (e.g., the times of arrival). The recording device may do so, for example, by storing and maintaining a clock indicating the current time. As the recording device receives the audio data of the first set of audio data, the recording device can label the audio data with timestamps indicating the times in which the recording device received the audio data or the times in which the recording device received the audio data minus a defined offset to account for the time it may take for the microphone to generate and transmit the audio data. The recording device can generate the spectrograms with the timestamps.

The recording device may generate spectrograms for sets of audio data from audio data generated by the microphone collected within defined time periods. For example, the recording device can sample data from different time periods into sets of audio data. The time periods may be overlapping or non-overlapping with each other. The time periods may have defined lengths. The recording device may segment the audio recordings into sets of audio data for the different time periods.

In step **204**, the recording device may execute a first machine learning model using the first set of audio data as input. The first machine learning model may be a neural network, a support vector machine, a random forest, etc., that is trained to analyze spectrograms to determine whether gunshots occurred during the time period of the spectrogram. The first machine learning model may be a binary classification machine learning model that outputs a binary value indicating that a spectrogram does include audio data of a gunshot or that a spectrogram does not include audio data of a gunshot. The recording device may execute the first machine learning model using the spectrogram of the first set of audio data as input. The first machine learning model

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may output a binary value indicating whether the first set of audio data includes audio data of a gunshot. If the first machine learning model outputs a value indicating the first set of audio data does not include audio data of a gunshot, the recording device and data processing system may stop performing the method 200. However, if the first machine learning model outputs a value indicating the first set of audio data does include audio data of a gunshot, the recording device may determine the first set of audio data is associated with a gunshot.

Responsive to determining the first set of audio data is associated with a gunshot, in step 206, the recording device may transmit the first set of audio data to the data processing system (e.g., a first remote processor). The recording device may transmit the first set of audio data to the data processing system as the audio data itself or as the spectrogram the recording device generated from the first set of audio data for processing. In some cases, the recording device may label the audio data to indicate the times in which the audio data was generated by the microphone or received by the recording device (e.g., the times of arrival) and transmit the labeled audio data to the data processing system. The recording device may repeat steps 202-206 for each set of audio data the recording device receives and/or spectrogram the recording device generates.

In step 208, the data processing system may receive the first set of audio data. The data processing system may additionally receive sets of audio data from other recording devices that are in communication with the data processing system that perform the same or similar steps to steps 202-206. The data processing system may receive the sets of audio data and store the sets of audio data in memory with stored associations with identifiers of the recording devices that transmitted the sets of audio data. The data processing system may determine sets of audio data may contain audio data of the same gunshot based on the sets of audio data having at least one timestamp for audio data in common or based on receipt of the sets of audio data within a time threshold of each other. The data processing system may determine sets of audio data may contain data of the same gunshot and analyze the sets audio data responsive to the determination.

In some cases, the data processing system may receive the first set of audio data and query recording devices that are near (e.g., within a distance threshold of) the recording device that transmitted the first set of audio data for further audio data. The data processing system may include a timestamp indicating a time of receipt of audio data or of an identified gunshot from the audio data (identified as described in the step 210) in the query. The queried recording devices may receive the query and perform the steps 202-206 with audio data that contains a time that matches the timestamp of the query to determine whether to transmit a set of such audio data to the data processing system.

In instances in which the data processing system receives a set of audio data instead of a spectrogram of audio data, the data processing system may generate a spectrogram from the set of audio data. The data processing system may generate the spectrogram with timestamps the recording device includes with the set of audio data (e.g., timestamps of the labeled set of audio data). The data processing system may generate the spectrogram using the same or similar techniques to the techniques the recording device generates spectrograms, as described with reference to step 202.

In step 210, the data processing system may iteratively execute a second machine learning model to determine a time of a gunshot for each set of audio data. The second

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machine learning model may be a neural network, a support vector machine, a random forest, etc., that is trained to analyze spectrograms to identify times of one or more gunshots from each individual spectrogram. The data processing system may execute the second machine learning model for each spectrogram that illustrates a set of audio data. The second machine learning model may output values identifying times (e.g., timestamps) of gunshots that the second machine learning models detects in the audio data of the spectrograms. The second machine learning model may identify the times from impulses in the spectrograms that correspond to the times. The data processing system may determine the output times of the second machine learning model are times of separate gunshots. The data processing system may use such times from the different spectrograms to determine a location of a gunshot or series of gunshots.

In some cases, the data processing system may select a time of a single gunshot from spectrograms from which the second machine learning model detected multiple gunshots. The data processing system may select the gunshot according to one or more gunshot selection rules. For example, the data processing system may store a rule that indicates to use the time of the first selected gunshot for each spectrogram the second machine learning model analyzes. In executing the rule, the data processing system may identify the earliest time stamp the second machine learning outputs for each spectrogram to use to detect the location of the gunshot. The data processing system may discard (e.g., remove from memory or from a cache) any other timestamps the second machine learning model output for the same spectrograms. The data processing system may select any number of times and/or times in any order (e.g., the second earliest time-stamp, the third earliest time stamp, etc.). In this way, the data processing system may identify a time in which a single gunshot occurred and avoid identifying times of echoes of the gunshot or of gunshots that rapidly occurred after the initial gunshot, which may be common when gunshots are detected from an automated weapon.

In some cases, the second machine learning model may output an indication that no gunshot was detected. In such cases, the data processing system may discard any indication that the recording device that transmitted the set of audio data and/or the spectrogram from memory.

In some cases, the data processing system may train the machine learning models stored on the recording devices based on the output of the second machine learning model at the data processing system. For example, responsive to the second machine learning model outputting an indication that a spectrogram does not include audio data for a gunshot, the data processing system may transmit an indication to the remote device that transmitted the spectrogram indicating that the spectrogram was inaccurately identified as being associated with a gunshot. Such false detections may occur, for example, when the recording devices record other sudden sounds such as a jackhammer, bus, or a fireworks. The remote computing device may receive the indication and use back-propagation techniques with a loss function to update the weights or parameters of the machine learning model that indicated the audio data or the spectrogram associated with the audio data is associated with a gunshot. The data processing system may operate to train machine learning models of any number of recording devices. In this way, the data processing system may tune each machine learning model to become more accurate over time and learn to ignore new sounds (e.g., new devices or vehicles that may emit or generate sounds that sound like gunshots but are not

gunshots) that are introduced into the environment or that are unique to the locations of the recording devices.

In some cases, prior to executing the second machine learning model to detect times of gunshots from the sets of audio data, the data processing system may execute a firework detection machine learning model using the sets of audio data to filter out spectrograms that include audio data of fireworks instead of gunshots. The firework detection machine learning model may be trained as a classification model that indicates whether the spectrograms include sounds of fireworks instead of a gunshot. For example, exploding fireworks may cause recording devices to generate audio data and/or spectrograms from audio data of the fireworks that is similar to the sounds of gunshots. The similarities in sound can cause the recording devices to generate a “false positive” and determine a spectrogram including sounds of fireworks exploding instead includes sounds of gunshots. These determinations can cause the recording devices to send audio data and/or spectrograms of the sound recordings of the exploding fireworks to the data processing system. To filter out spectrograms and/or audio data of exploding fireworks, the data processing system may execute the firework detection machine learning model using the spectrograms for the sets of audio data (e.g., spectrograms that the recording devices send the data processing system and/or spectrograms that the data processing system generates from received audio data) as input. The firework detection machine learning model may output indications of whether the spectrograms are spectrograms of audio data for fireworks or spectrograms of audio data for gunshots. For each spectrogram for which the firework detection machine learning model outputs an indication the spectrogram is of fireworks, the firework detection machine learning model may discard (e.g., remove from memory or a cache of the data processing system) or otherwise flag the spectrogram as being associated with fireworks. The data processing system may not use such spectrograms (e.g., the data processing system may identify the flags associated with the spectrograms and not use the spectrograms based on the identifications) as input into the second machine learning model when determining times of gunshots to minimize processing resources of detecting gunshots and avoid false identifications of gunshots.

The data processing system may additionally transmit an indication to the recording device that transmitted the flagged or discarded spectrogram indicating the recording devices determination that the audio data contained audio data of a gunshot was incorrect. The recording device may receive the indication and train the machine learning model at the recording device using back-propagations techniques based on the indication. Accordingly, the data processing system continuously update the normative data model for data captured at each sensor site.

In some cases, instead of the data processing system filtering out spectrograms based on whether the spectrograms contain audio data of fireworks, the recording devices can locally filter out such spectrograms. For example, the recording devices may each store a firework detection machine learning model similar to the firework detection machine learning model described above. Upon determining a spectrogram is associated with a gunshot, a recording device may execute a firework detection machine learning model with the spectrogram to determine if the determination was a false positive and/or if the spectrogram contains audio data of fireworks instead of a gunshot. If the firework detection machine learning model outputs an indication that the spectrogram is a spectrogram for a gunshot, the record-

ing device may discard the spectrogram and not send the spectrogram or the audio data associated with the spectrogram to the data processing system. Otherwise, the firework detection machine learning model may transmit the spectrogram or the audio data associated with the spectrogram to the data processing system. In this way, the recording devices may avoid sending spectrograms and/or audio data of fireworks to the data processing system, which can lower the bandwidth requirements and/or processing requirements of detecting gunshots at the data processing system.

In step 212, the data processing system may execute a multilateration model using the time of the gunshot for each set of audio data. The data processing system may do so to determine a location of a gunshot. For example, the data processing system may generate a list of groupings of three recording devices from which the data processing system received a set of audio data and/or a spectrogram that the second machine learning model identified at least one time of a gunshot. To generate the list, the data processing system may retrieve the identifications of the recording devices that transmitted the sets of audio data and/or spectrograms from memory and calculate different permutations of three of the recording devices. The data processing system may generate the list as a data structure that includes each permutation of three recording devices and the timestamps of the times in which gunshots that were detected and/or that were selected from the audio data or spectrograms the recording devices transmitted to the data processing system.

The data processing system may execute the multilateration model to cause the multilateration model to calculate a potential location of the gunshot for each grouping of three recording devices. The multilateration model may do so, for example, by applying the following set of equations:

$$(x_s - x_1)^2 + (y_s - y_1)^2 = (r_s - s_1 r_1)^2$$

$$(x_s - x_2)^2 + (y_s - y_2)^2 = (r_s - s_2 r_2)^2$$

$$(x_s - x_3)^2 + (y_s - y_3)^2 = (r_s - s_3 r_3)^2$$

where r is the time difference of arrival, x and y are the coordinates (e.g., the geographic locations), and s is the solution for the variable. The detected or selected times of the gunshots and the locations of the recording devices in the groupings of three recording devices may be used as input into the equations.

Upon calculating the potential locations, the multilateration model may iteratively filter out the different groupings of recording devices. For example, the multilateration model may calculate an average gunshot location from the calculated potential gunshot locations of each grouping of three recording devices that detected the gunshot. The multilateration model may then use a distance formula between the potential locations of the gunshot for each grouping of recording devices to calculate distances between the potential locations of the groupings and the average location of all of the groupings. The multilateration model may identify the grouping or a set number of groupings associated with the largest distance or distances and remove the groupings from the list. In some cases, the multilateration model may remove any groupings from the list that are associated with a distance that exceeds a threshold. The multilateration model may then calculate a new average location from the potential locations of the remaining recording devices on the list. The multilateration model may calculate distances between the potential locations and the new average location and remove any groupings of recording devices that are associated with a largest calculated distance or a distance

that exceeds a threshold. The multilateration model may repeatedly perform this process until only one grouping of recording devices is left on the list. The multilateration model may identify the potential location of the remaining grouping as the location of the gunshot.

In step **214**, the data processing system may transmit an indication of the location to a remote processor (e.g., a second remote processor) of a second remote computing device. The data processing system may transmit the indication of the location by generating a record (e.g., a file, document, table, listing, message, notification, etc.) that includes the indication of the location. The data processing system may transmit the indication of the location as a string identifying the location. In some cases the data processing system may additionally or instead include a timestamp indicating a time in which one of the three remote computing devices generated audio data for the gunshot. The data processing system may select a time for one (e.g., the earliest time) or all three of the remote computing devices and insert the selected time or times in the record. The data processing system may transmit the record to the remote processor.

Transmitting the indication of the location to the remote processor may cause the remote processor to direct an unmanned vehicle to the location of the gunshot. For example, the remote processor may be configured to control a drone with an attached camera or video camera. The remote processor may receive the location in the record from the data processing system and configure the drone to travel to the location. The drone may drive or fly to the location and capture video or pictures of the location. By doing so, the system may operate to capture evidence or images of the site such that investigators may better piece together the story of why the gunshot was fired and/or who was involved in firing the gunshot.

In one example, the data processing system can control a camera to rotate in the direction of the location of the gunshot. For example, the data processing system can store the locations of different cameras around a municipal area. In some cases, the data processing system can store field of view (e.g., field of view) radii of the different cameras. Upon determining the location of the gunshot, the data processing system can determine if the location is within the field of view of any of the cameras. The data processing system can instead identify the camera that is closest to the location (e.g., identify the location and use a distance function between the location and the locations of the cameras to identify the camera that is closest to the location). The data processing system can determine a location of the gunshot relative to the identified camera based on the geographic data of the location of the gunshot and the geographic location of the camera. The data processing system can control the camera to rotate to the location of the gunshot (e.g., the data processing system can transmit an azimuth, an elevation, and/or global positioning system coordinates to the camera to cause the camera to rotate to the location of the gunshot).

In another example, the data processing system can control a camera to rotate towards a location of a lane of egress. The lanes of egress may be escape routes of individuals who committed a crime, for example. The lanes of egress may be predetermined locations and/or distances relative to a location of a gunshot (e.g., 50 yards away from a location of a gunshot or a predetermined or stored location away from a location of a gunshot). Locations of lanes of egress can be stored in memory of the data processing system as specific geographical locations or functions to determine the geo-

graphic locations of the lanes of egress (e.g., identify the geographic location of a lane of egress by aggregating 50 yards in the x direction from the location of the gunshot). The data processing system can retrieve a location of a lane of egress by querying memory using the location of the gunshot as an index. Upon retrieving the location from memory, the data processing system can identify a camera that corresponds to the location of the gunshot (e.g., that has a stored association with a radius that includes the location of the gunshot) and/or the location of the lane of egress (e.g., that has a stored association with a radius that includes the location of the lane of egress). The data processing system can determine the location of the lane of egress by executing a stored function based on the location of the gunshot. The data processing system can control the camera to rotate towards the location of the lane of egress. Accordingly, the data processing system can control the camera to capture video of potential escape routes of individuals that may have just committed a crime.

FIG. 3 is a sequence diagram of a sequence **300** for gunshot detection, according to an embodiment. The sequence **300** may be performed by any number of components, such as components of the gunshot detection system **100**, shown and described with reference to FIG. 1. In the sequence **300**, recording devices **302**, **304**, and **306** (e.g., the recording devices **120**) may operate together to generate audio data of sound waves from the surrounding environment (e.g., a metropolitan area). The recording devices **302**, **304**, and **306** may identify sets of audio data that contain audio of gunshots and transmit the audio data (e.g., transmit spectrograms of the audio data) to an update server **308**, an orchestration server **310**, an IP address management (IPAM)/monitoring server **312**, and/or an artificial intelligence/multilateration server **314**. Together, the orchestration server **310**, the IPAM/monitoring server **312**, and/or the artificial intelligence/multilateration server **314** may be or include the analytics server **110a**. The recording devices **302**, **304**, and **306** may transmit the audio data and/or spectrograms through an access point **317** and a virtual private network (VPN) **318**. The update server **308**, orchestration server **310**, IPAM/monitoring server **312**, and/or artificial intelligence/multilateration server **314** may operate together to perform the method **200** to detect locations and/or times of gunshots around a particular area from audio data and/or spectrograms the update server **308**, orchestration server **310**, IPAM/monitoring server **312**, and/or artificial intelligence/multilateration server **314** receive from the recording devices **302**, **304**, and **306**.

An operator can access the orchestration server **310**, the IPAM/monitoring server **312**, and/or the artificial intelligence/multilateration server **314** via an operator graphical user interface **316**. In operating the servers **310**, **312**, and **314**, the operator can configure the servers **310**, **312**, and **314** to operate to detect gunshots. For instance, the operator can change the configuration of the artificial intelligence model (e.g., the second machine learning model and/or the firework detection machine learning model) and/or a multilateration model of the artificial intelligence/multilateration server **314** to select specific potential gunshots from spectrograms to use for multilateration. The graphical user interface **316** may illustrate the locations of gunshots on a map with other relevant information (e.g., the time of the gunshot). The graphical user interface **316** may integrate with emergency response software such that a user may operate the graphical user interface **316** to transmit a message to a server of an emergency response team (e.g., authorities) indicating the gunshot occurred and/or the location of the gunshot.

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The artificial intelligence/multilateration server **314** may identify specific times of the gunshots from impulses in spectrograms. The artificial intelligence/multilateration server **314** may be configured to identify coordinates of the recording devices **302-306** and the corresponding times of arrival of the gunshot sounds the recording devices record and output a location of gunshot using a multilateration model based on the coordinates and the times of arrival.

The IPAM/monitoring server **312** may control how often the IPAM/monitoring server **312** requests data from the recording devices **302-306**. The IPAM/monitoring server **312** may also store of the locations of the recording devices **302-306**. In one example, the IPAM/monitoring server **312** may store geolocation data of the recording devices **302-306**.

The code **318** and the continuous integration build server **320** may operate to continuously update the update server **308** and the orchestration server **310** for performance maintenance and improvements. For example, the code **318** may be configured or written by a programmer and the continuous integration build server **320** may propagate updates through the update server **308** to ensure the recording devices **302-306** are configured with the same settings. The orchestration server **310** may provision servers so the server can be reliably burned down and rebuilt. Additionally, the orchestration server **310** can interact with the recording devices **302-306** individually or together.

In some cases, the update server **308** may store a copy of the machine learning model that is stored locally at the recording devices **302-306** and train the copy of the machine learning model over time. The update server **308** may do so, for example, by training the copy with new sounds of new devices (e.g., a jackhammer) that may become available or located at the locations of the recording devices **302-306**. The update server **308** may train the machine learning model and transmit files (e.g., binary files) containing the machine learning model to the recording devices to update the machine learning models that are stored locally at the recording devices **302-306**.

The recording devices **302-306** may be tagged (e.g., geotagged) using RTK geotagging techniques. For example, an RTK geotagging handheld device **322** may communicate with an RTK geotagging basestation **324** to detect the location of the recording devices **302-306**. The RTK geotagging handheld device **322** and the RTK geotagging basestation **324** may do so by assigning locations to the recording devices **302-306** upon installation. Barcodes (e.g., substantially weatherproof stickers containing barcodes) may be attached to the recording devices **302-306**. The RTK geotagging handheld device **322** may scan the barcodes and enter the barcodes into a database. The RTK geotagging handheld device **322** may communicate with the RTK geotagging basestation **324** to determine the current location (e.g., coordinates) of the RTK geotagging handheld device **322** and, thus, the respective recording devices **302-306**. The RTK geotagging handheld device **322** may store the locations of the recording devices **302-306** in memory and/or in memory of the IPAM/monitoring server **312** such that the locations may be retrieved and used to locate gunshots as described herein.

FIG. 4 is a sequence diagram of a sequence **400** for gunshot detection, according to an embodiment. The sequence **400** may be performed by any number of components, such as the components of the gunshot detection system **100**, shown and described with reference to FIG. 1 or 3. For example, in the sequence **400**, a recording device **402** (e.g., a recording device **120**, or one of the recording

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devices **302, 304, and 306**) can operate to record sounds of the environment surrounding the recording device **402**. The recording device **402** may transmit pulses in a pulse per second (PPS) configuration **404** to every device of the gunshot detection system **100** and/or described with reference to the sequence **300**.

A microphone **406** may include one or more microphones that capture sound waves from the environment and convert the soundwaves to audio data. The microphone **406** may transmit the audio data to an advanced Advanced Linux Sound Architecture (ALSA) driver **408**. The ALSA driver **408** may increase the gain on the microphone **406**. A ring buffer **410** of the recording device **402** may continually generate audio samples (e.g., sets of audio data) with or without overlap from the audio data the ring buffer receives from the ALSA driver **408**. In cases in which there is overlap between the samples, there may be a trade off between response time and processing load because the audio data of the overlap is processed twice. The ring buffer **410** may additionally timestamp the audio to indicate the times in which the audio was received (e.g., times of arrival). The timestamps may be used for artificial intelligence classification to detect gunshots from the audio as described herein.

An artificial intelligence (AI) model **412** of the recording device **402** may analyze the sample sets of data that the ring buffer **410** generates. To do so, an AI preselector **414** of the recording device **402** may preprocess the samples of audio data into discrete spectrograms such that the audio data can be processed with image processing techniques. The AI model **412** may use image processing techniques on the spectrograms the AI preselector **414** generated or otherwise received to detect spectrograms that may include audio data of gunshots. In doing so, the AI model **412** may determine identifications **416** of whether the spectrograms include audio data of gunshots or not. The recording device **402** may transmit, through a VPN client **418** and over a network **420**, spectrograms and/or audio data that was used to generate such spectrograms to a server (e.g., the analytics server **110a** or the artificial intelligence/multilateration server **314**) that the recording device **402** determined are associated with gunshots.

The server may query nearby recording devices (e.g., recording devices within a threshold distance of the recording device **402**) for sets of audio data and/or spectrograms of such sets of audio data that were generated concurrently or within a threshold (e.g., a defined time threshold) of a time (e.g., a beginning, end time, or a time between the beginning and end time) of the set of audio data from which the gunshot was originally detected at the recording device **402**. The server may do so by including an timestamp of the time in query. In response to the query, the queried recording devices may identify sets of audio data and/or spectrograms of sets of audio data that contain audio from the time and transmit the sets of audio data and/or spectrograms to the server for gunshot detection from each set of audio data and/or spectrogram. In some cases, the queried recording devices may first determine which of the spectrograms or sets of audio data contain audio data of a gunshot and only transmit spectrograms and/or sets of audio data the recording devices determined are associated with gunshots. In some cases, the queried recording devices may transmit all sets of audio data and/or recording devices that contain audio data from the time in the query.

The recording device **402** may communicate with other recording devices and/or the network using a variety of communication techniques. For example, the recording device **402** may include a cell modem **422** that connects the

recording device **402** to the network **420**. In another example, the recording device **402** may include a wireless communication interface **424**, such as 802.11ah. The recording device **402** may transmit and receive messages from other recording devices via the wireless communication interface **424**. For instance, the recording device **402** may receive audio data and/or spectrograms from other recording devices of a mesh network of which the recording device **402** is a part. The recording device **402** may transmit the received spectrogram and/or audio data to a server for processing via the cell modem **422** and through the network **420**.

The recording device **402** may additionally include other components. For instance, the recording device **402** may include a GPS **426**, a GPSD **428**, an NTPD **430**, a monitoring client **432**, a watchdog circuit **434**, an RTK geotag **436**, an update client **438**, and/or a preempt-RT **440**. The GPS **426** may be a global positioning system that enables the recording device **402** to determine a current time at the location of the recording device **402**. The GPS **426** may transmit the determined current time to the GPSD **428**, which can communicate the data to a server (e.g., the analytics server **110a**) to indicate the current time. The NTPD **430** may be an operating system daemon which synchronizes the time and day with a time and day stored on a network. The recording device **402** may retrieve the synchronized time and data for samples of audio data to label the audio data. The monitoring client **432** may communicate with a monitoring server (e.g., may be an application programming interface that enables communication with the monitoring server), enabling the monitoring server to communicate with and/or retrieve data from the recording device **402**. The watchdog circuit **434** may automatically reset the processor of the recording device **402** upon determining the processor is unresponsive. The watchdog circuit **434** may operate in the background. The watchdog circuit **434** may perform a hard power cycle the recording device **402** if the recording device **402** becomes unreachable. In some cases, three reboots without successfully establishing a connection with the network **420** may cause the watchdog circuit **434** to revert the recording device **402** to a previous update. The RTK geotagging **436** may be a geographical tag identifying the geographical location of the recording device **402** (which may be determined upon installation of the recording device **402** based on GPS data of an RTK tagging device that is present during installation). The recording device **402** may store the geographical tag of the RTK geotagging **436** in memory. The update client **438** may be an application programming interface (API) that facilitates communication with an update server to receive updates for the software or firmware of the recording device **402**. The preempt-RT **440** may enable the recording device **402** to run special real-time tasks and interrupt handlers. The preempt-RT **440** may further make interrupts as threads.

FIG. **5** is an illustration of a network mesh **500** for a gunshot detection system, according to an embodiment. The network mesh **500** may include multiple recording devices **502a-g** (collectively, recording devices **502** and individually, recording device **502**) (e.g., the recording devices **120**) that are distributed around a metropolitan area. The recording devices **502** in the network mesh **500** may communicate with each other (e.g., communicate via 802.11ah). A recording device of the recording devices **502** can include a modem (e.g., a cell modem) that connects the recording device to a network **504**. Any number of the recording devices can include such a modem. The recording devices **502** can transmit messages to each other in paths to reach a

recording device that includes a modem or a modem that has an established connection with a server through the network **504**. The recording device including such a modem may transmit the received data to the server for further processing. For example, upon determining audio data includes audio data of a gunshot, a recording device can transmit the audio data through a sequence of recording devices to reach a recording device with a modem so the recording device with the modem can transmit the audio data to a server to determine the time of the gunshot, as described herein.

In some cases, each of the recording devices **502** may include both a modem to directly connect to the network **504** and a Wi-Fi interface to communicate with each other. To minimize the number of recording devices **502** that communicate through the network **504**, only one recording device **502** may operate a modem and the other recording devices **502** may communicate with the one recording device **502** to transmit audio data and/or spectrograms through the network **504**. However, if a recording device **502** is not able to establish a connection with any other recording devices **502**, the recording device **502** may activate the modem within the recording device **502** to communicate directly through the network **504** (e.g., transmit audio data and/or spectrograms directly to the network).

FIG. **6** is an illustration **600** of locations of recording devices of a gunshot detection system within a metropolitan area (e.g., a city), according to an embodiment. The illustration **600** may be a graphical user interface presented on a screen of a computing device. Multiple locations **602a-f** (together locations **602**) of recording devices (e.g., recording devices **120**) are marked in the illustration **600**. The illustration **600** also includes potential gunshot locations **604** of a gunshot for various permutations of the recording devices that generated audio data for the gunshot. The potential gunshot locations **604** may be calculated based on times the recording devices generated audio data of the gunshot and/or the locations **602** of the recording devices. The potential gunshot locations **604** may be used in a multilateration technique to calculate or determine the actual location of the gunshot.

FIG. **7** illustrates a depictions **700** of a set of audio data (e.g., a sample of audio data), according to an embodiment. The depictions **700** may include a beat waveform **702** and a spectrogram **704**. The beat waveform **702** may illustrate a detected soundwave over time. The spectrogram **704** may illustrate frequencies of the detected soundwave over time. The machine learning models described herein may use image recognition techniques on illustrations such as the beat waveform **702** and the spectrogram **704** to detect, in the case of recording devices (e.g., recording devices **120**), whether the illustrations illustrate audio data for at least one gunshot, and/or, in the case of a server (e.g., the analytics server **110a**) times of one or more gunshots. References to generating and processing spectrograms herein may be references to generating and processing other visual representations of sound waves, such as the beat waveform **702**.

FIG. **8** illustrates a housing **800** for a recording device, according to an embodiment. The housing **800** can be a housing for a recording device (e.g., the recording device **120**). The housing **800** may be configured to removably couple with light fixtures (e.g., light fixtures attached or coupled with light posts (e.g., light posts **125**)) around a metropolitan area. In one example, the housing **800** can be configured such that a microphone, a communications interface, and/or a processor can be mounted in or on the housing **800**. The housing **800** can also be configured such that the housing **800** can be coupled to or with a light post. For

example, the housing **800** may use a twist lock plug and socket system into an opening or receptacle of a light post. Because of this configuration, the housing **800** (and thus the recording device itself) may be quickly installed throughout a metropolitan area to existing light posts instead of requiring construction of new structures that can hold each recording device.

The housing **800** may include a top enclosure **802** and a bottom enclosure **804**. The top enclosure **802** and the bottom enclosure **804** can house a printed circuit board (PCB) **806**. The PCB **806** can include a processor or other computer that is configured to receive and process audio data from a microphone. An antenna **808**, a thermal switch **810**, a heater mounting plate **812**, and/or a positive temperature coefficient (PTC) heater **814** can be mounted to the PCB **806**. The PCB **806** can be mounted to the top enclosure **802** and the bottom enclosure **804** with posts **816**, a three dimensional printed bracket **818**, and an O-ring **820**. A WI-FI mesh **822** may also be coupled to the top enclosure **802** and/or the bottom enclosure **804**. The PCB **806** can be mounted between the top enclosure **802** and the bottom enclosure **804** by screws **824** or other fasteners. Additionally, a plug **826** can be mounted or attached to the bottom enclosure **804** with screws **828**. The plug **826** may connect the electrical components inside the housing **800** with a light fixture (e.g., a light fixture attached or couple to a light post) to power the electrical components. Additionally, a receptacle **830** can be mounted or attached to the top enclosure **802** with screws **832**. The receptacle **830** can be configured to hold or store a light (e.g., a photoelectric lightbulb) and may enable the light to connect with the plug **826**. Accordingly, during installation, an operator may replace a light bulb in a light post with the housing **800** and place the light in the receptacle **830** to provide the light post with sound detection capabilities in addition to lighting capabilities. The plug **826** of the bottom enclosure **804** can be configured to be fastened to a light fixture as a twist-lock fastening system.

The plug **826** may be configured to physically match plugs of photocells that are connected to light fixtures. For example, the plug **826** may have a twist-lock fastening configuration similar to a photocell that can fasten the housing **800** to a light fixture in place of the photocell. The top enclosure **802** may be configured to receive photocells with configurations similar to the plug **826** (e.g., receive the photocell at an opposite end of the housing **800**, such as at an external face of the top enclosure **802**). Accordingly, when the housing **800** is deployed to be attached to light fixtures, a user may unplug a photocell from a light fixture and replace the photocell with the plug **826**. The user may then plug in the replaced photocell to the receptacle **830** on top of the top enclosure **802**. In this way, the housing **800** and the electronics enclosed within the housing **800** may be coupled with light fixtures using the existing structure of the light fixtures, enabling faster and easier installation compared with enclosures that require construction of new structures.

The housing **800** can include a series of tabs extending from a bottom of the bottom enclosure **804**. Once the housing **800** is installed above a light fixture, a safety cable can be threaded through holes in the tabs and around the arm supporting the light fixture in a manner that can allow the safety cable to keep tension on a twist-lock plug engagement to the socket on the top of the light fixture. This configuration can serve to keep the housing **800** in place as well as to prevent the housing **800** from falling to the ground should the housing **800** become separated from the light fixture.

The housing **800** can configured to have environmental protections. For example, the outer edge of the housing **800** can form a drip edge. The drip edge can limit water flow across the bottom of the housing **800** which could detract from the sensing abilities as well as slip through an electrical seal at the fixture connection based on persistent exposure. An extended ring/ridge around the sensor (e.g., microphone) inside or mounted to the housing **800** can create another drip edge and barrier protecting the opening to the sensor. A ring extending from the bottom enclosure **804** around the plug **826** (e.g., the twist-lock plug) can serve as another barrier to water infiltration into the electrical connection. Specialized membranes designed to protect the sensor (microphone) from water but allow the passage of a gas with negligible degradation of the acoustic waveform can cover the entrance to the housings of both devices. The top enclosure **802** and the bottom enclosure **804** can be designed to provide advanced weather resistance by protecting the sealing surfaces from direct exposure to driven rain and spray. The O-ring **820** can also be used to seal the enclosures **802** and **804**.

The housing **800** can be configured to improve audio detection by the sensors (e.g., microphone **406**). For example, the housing **800** can include specialized isolation using micro-fine, ultra-flexible wiring and floss cocooning of the sensors to minimize transfer of mechanical noise to the sensor. The specialized isolation can reduce the need for digital noise reduction techniques which can degrade spectrographic images. Additionally, the conical shape of the microphone opening in the housing **800** can serve to amplify/intensify the pressure wave and focus it on the sensor above. Using a design based on the exponential spherisches curve, the housing **800** can be configured to have an acoustic geometry that creates a natural amplification technique.

Advantageously, by being configured in this manner, the housing **800** can be installed to a fused electrical disconnect of a utility or another device (e.g., a side or front of a building). The configuration of the housing **800** can help avoid the process of installing a fused electrical disconnect to a utility pole, have the local power authority connect power to the disconnect, mount a box to the utility pole, connect the box to the electrical disconnect, and mount the detection inside of the weatherproof box. Deployment in this manner can take multiple days to effect and many hours to deploy. Deployment of the housing **800** can instead take minutes to remove a photocell from a receptacle of a light fixture, replace the photocell with the housing **800** by installing the housing **800** into the receptacle of the light fixture, and insert the photocell into a receptacle on top of the housing **800**. This configuration can provide power to the housing **800** from the receptacle (e.g., the socket) of the light fixture, which can then provide power to the photocell. The photocell can pass electrical signals (e.g., electrical signals to control the light of the light fixture) through the housing **800** back to the light fixture. Accordingly, the housing **800** can be installed into the light fixture to allow the light fixture to continue operation of turning the light of the light fixture on when the sun sets and turning the light off at sunrise.

FIG. 9 illustrates perspective view **900** of the top enclosure **802**, according to an embodiment. As illustrated, the receptacle **830** is mounted to the top enclosure **802**.

FIG. 10 illustrates a side view **1000** of the housing **800**, according to an embodiment. In the side view **1000**, the top enclosure **802** and the bottom enclosure **804** may be coupled together by screws or other fasteners. Mounting brackets **1002** may coupled or fastened to a bottom surface **1004** of the bottom enclosure **804**.

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FIG. 11 illustrates a cross-sectional view 1100 of the housing 800, according to an embodiment. In the cross-sectional view 1100, the receptacle 830 may be coupled with the top enclosure 802 by the screws 832. The bottom enclosure 804 may be coupled with the top enclosure 802 by the screws 824.

FIG. 12 illustrates a top view 1200 of the printed circuit board 806, according to an embodiment.

FIG. 13 illustrates a top view 1300 of a top surface 1302 of the bottom enclosure 804, according to an embodiment.

FIG. 14 illustrates a bottom view 1400 of a bottom surface 1402 of the top enclosure 802, according to an embodiment.

FIG. 15 illustrates a perspective view 1500 of the PCB 806, according to an embodiment. The PCB 806 may include the circuitry of a recording device including a microphone and a processor that receives and processes audio data from the microphone. The processor may use machine learning techniques to detect sets of audio data that contain audio of gunshots.

FIGS. 16A-C illustrate perspective views 1600-1604 of different light fixtures coupled to light posts to which a recording device (e.g., the recording device 120) can be coupled, according to an embodiment. A light fixture 1606 can be coupled to a light post 1608. In the perspective view 1602, a light fixture 1610 can be coupled with a light post 1612. In the perspective view 1604, a light fixture 1614 can be coupled with a light post 1616. The perspective view 1604 can also include a light fixture 1618 coupled with a building 1620. Each light fixture 1606, 1610, and 1614 can receive power through power cords running through the light posts 1608, 1612, and 1616 to which the light fixtures 1606, 1610, and 1614 are attached. Lights can be coupled or attached to the light fixtures 1606, 1610, and 1614 to receive power and light up. Light fixtures can be coupled to lights of any configuration.

While light fixtures as described herein are described as coupled to different light posts, light fixtures can be coupled to other objects through power cords as well. For example, as illustrated in the perspective view 1604, the light fixture 1618 can be coupled with the building 1620. The building 1620 can be electrically coupled to a power or utility grid and the light fixture 1618 can be coupled to the same power or utility grid through wiring of the building 1620.

Recording devices (e.g., the recording device 120) can be coupled to the light fixtures 1606, 1610, and 1614. Recording devices can be coupled to the light fixtures 1606, 1610, and 1614 using a male and female twist lock electrical connector between the housings of the recording devices and the light fixtures 1606, 1610, and 1614. In doings, photocells that were original connected to the light fixtures 1606, 1610, and 1614 can be removed from the light fixtures 1606, 1610, and 1614 and replaced with the recording devices. The photocells can then be placed onto opposing ends of the housings of the recording devices from the ends that have the twist-lock electrical connection with the light fixtures 1606, 1610, and 1614. In instances in which a light post does not have a light fixture but includes a live twist-lock socket, a recording device can be directly coupled or connected to the light post through the live twist-lock socket.

FIG. 17 illustrates a perspective view 1700 of a recording device 1702, according to an embodiment. The recording device 1702 may be the same as or similar to the recording device 120. The recording device 1702 may include a housing with a top enclosure 1704 and a bottom enclosure 1706. A PCB 1708 may be mounted or attached to the bottom enclosure 1706. The PCB 1708 may have power cords 1710 plugged into circuitry mounted on the PCB 1708.

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The PCB 1708 may include or be connected to an array of microphones (e.g., multiple microphones connected to the PCB 1708). The microphones may be the same as or similar to the microphones 406. The power cords 1710 may enable a camera to mount or attach to the top enclosure 1704, such as in a receptacle of the top enclosure 1704. In some cases, the power cords 1710 may connect the circuitry of the PCB 1708 to the camera mounted to or on top of the top enclosure 1704. The power cords 1710 may provide power to the camera and/or enable the circuitry of the PCB 1708 to rotate the camera to locations of detected gunshots and otherwise control the camera. In some cases, the circuitry may include a communications interface to communicate with a camera that is remote from the recording device 1702 over a network.

A housing of the recording device 1702 can screw to the bottom of a mounting surface of an enclosure (e.g., a housing of another recording device, such as the recording device 120, or a housing of a camera) or a mount on a surface. The recording device 1702 can include a processor coupled to circuitry of the PCB 1708. The processor can be embedded into a sensor array board (e.g., a board containing an array of sensors or microphones) of the recording device 1702, in some cases with a plurality of sensors. The sensors of the array of sensors may be the same as or similar to the microphone 406. The sensor array board may be a part of the PCB 1708. In some cases, the array of sensors may be separately dispersed around the PCB 1708. The processor can be powered by a power over Ethernet system, which can be the same system powering a camera (e.g., pan tilt zoom (PTZ) camera) the processor can control based on audio data generated by the sensor array.

The housing of the recording device 1702 can employ a self-tensioning closure methodology. The self-tensioning closure methodology can allow the top and bottom enclosures 1704 and 1706 to increase pressure onto an o-ring seal as the o-ring seal is twisted to a closed position. Detents, slots/grooves, and spring elements, created of the same or a similar material as the housing, can be molded as part of the housing. The detents can slide down the grooves as the cover is twisted and come to rest at the end of the groove, held in place by tension and the synthetic spring created by the unique design. This housing can be created in a three-dimensional printed model, injection molded, or fabricated in other manners.

The processor can use audio data from the array of sensors or microphones to calculate the time of arrival of audio data generated actionable sounds (e.g., gunshots, car crashes, screams, etc.). Actionable sounds can be events as described above. For example, the processor can identify sets of audio data from the different microphones that contain audio data of an actionable sound. The processor can identify such sets of audio data by generating spectrograms from the sets of audio data generated by the different microphones and using machine learning techniques (e.g., machine learning techniques similar to those described herein such as those used by the first and/or second machine learning models described with reference to FIG. 2) on the spectrograms to identify sets of audio data that contain audio data of an actionable sound and/or times of the actionable sound in the different sets of audio data. The times of the actionable sounds may be times of arrival of the actionable sounds for the different sensors.

The processor can identify the location of the actionable sound relative to the camera. For example, based on the times of arrival and locations of the sensors that generated and/or provided audio data of the actionable sound, the

processor can use sound localization techniques to identify the location (e.g., the source) of the actionable sound relative to the array of sensors. The processor can then determine a location of the actionable sound relative to the camera using one or more algorithms, such as the law of cosines or the law of sines based on a vector from the location of the array of sensors to the location of the actionable sound, a vector from the location of the array of sensors to a location of the camera, and an angle between the two vectors. From this calculation, the processor can create a three-dimensional digital representation of the source of the sound in relevance to the location of the sensor array. That data can be modified algorithmically using the relative positioning of the sensor array and the zero position of the PTZ camera. The processor can generate a directional command including an azimuth and an elevation to rotate the camera to point to the location of the actionable sound. The processor can transmit the directional command to the camera, in some cases with an instruction to change the state of the camera to an "on" state to begin recording. The camera can receive the directional command and/or the instruction to change state and operate according to the command and instruction.

The processor can transmit an indication to a remote processor that indicates the actionable sound was detected and/or that the camera rotated to have a field of view including the location of the actionable sound. The processor can include an identification of the camera that was rotated and/or an identification of the recording device that detected the actionable sound in the indication. The remote processor may generate and present a user interface at a display of a remote computing device. Accordingly, a user or operator accessing the remote computing device can be notified in real-time that a camera has relevant live imagery for their review and/or that the camera field of view has been repositioned.

FIG. 18 illustrates a perspective view 1800 of the top enclosure 1704 of the recording device 1702, according to an embodiment.

FIG. 19 illustrates a perspective view 1900 of the bottom enclosure 1706 of the recording device 1702, according to an embodiment.

FIG. 20 is a flow diagram illustrating an exemplary method 2000 for detecting actionable sounds, according to an embodiment. The method 2000 may include steps 2002-2008. However, other embodiments may include additional or alternative steps, or may omit one or more steps altogether. The method 2000 is described as being executed by a recording device (e.g., one or both of the recording device 120 or the recording device 1702). However, one or more steps of the method 2000 may be executed by any number of computing devices operating in the distributed computing system. For instance, one or more computing devices may locally perform part or all of the steps described in FIG. 20 or a cloud device may perform such steps. The method 2000 can be performed in addition to or during any of the steps of the method 200.

In step 2002, the recording device may receive a set of audio data. A processor of the recording device can receive the set of audio data. The recording device can receive sets of audio data from each of a plurality of microphones. The plurality of microphones may be inside or mounted to a housing of the recording device. The sets of audio data may be or include a sound recording from audio collected within a particular time frame. The plurality of microphones may continuously record sounds made in the environment surrounding the recording device. As the microphones record

the sounds, the microphones can transmit the recorded sounds to the processor of the recording device for further processing.

The recording device may generate spectrograms from the sets of audio data. The recording device may generate the spectrograms for each of the sets of audio data, for example, by calculating the Fourier transforms of various segments (e.g., equal-length segments) of the set of audio data in which the frequency of the audio data does not substantially change. The recording device may then generate a graph of the output of the Fourier transform calculation. The graph may illustrate the decibels of the audio data at different frequencies.

When generating the spectrograms for the sets of audio data, the recording device may include timestamps in the spectrograms of the real-world times in which the respective audio data was generated or received from the plurality of microphones (e.g., the times of arrival). The recording device may do so, for example, by storing and maintaining a clock indicating the current time. As the recording device receives the audio data of the sets of audio data, the recording device can label the audio data with timestamps indicating the times in which the recording device received the audio data or the times in which the recording device received the audio data minus a defined offset to account for the time it may take for the microphones to generate and transmit the sets of audio data. The recording device can generate the spectrograms with the timestamps.

The recording device may generate spectrograms for sets of audio data from audio data generated by the microphones collected within defined time periods. For example, the recording device can sample data from different time periods into sets of audio data. The time periods may be overlapping or non-overlapping with each other. The time periods may have defined lengths. The recording device may segment the audio recordings into sets of audio data for the different time periods.

The recording device can generate separate spectrograms for sets of audio data the recording device generates or receives from each of the plurality of microphones. For example, during a time period, the recording device can concurrently receive audio data generated by the plurality of microphones within the housing of the recording device. The recording device can partition the audio data into sets of audio data that each correspond to the microphone from which the recording device received the audio data. The recording device can do so, for example, by labeling the audio data with identifiers (e.g., numeric or alphanumeric identifiers) of the microphones from which the recording device received the audio data. The recording device can receive the audio data from the different microphones and sample the audio data into different sets of audio data. The recording device can then generate spectrograms from the different sets of audio data.

In step 2004, the recording device may execute a machine learning model using each of the sets of audio data as input. The machine learning model may be a neural network, a deep learning model, a support vector machine, a random forest, etc., that is trained to analyze spectrograms to determine whether an actionable sound (e.g., a gunshot, a car crash, a scream, etc.) occurred during the time period of the spectrogram and/or a time of the actionable sound. The machine learning model may be a binary classification machine learning model that outputs a binary value indicating that a spectrogram does include audio data of an actionable sound or that a spectrogram does not include audio data of an actionable sound. The machine learning model may

additionally or instead output a time of the spectrogram in which the actionable sound occurred. The recording device may execute the machine learning model using the spectrogram of a set of audio data as input. The machine learning model may output a binary value indicating whether the set of audio data includes audio data of an actionable sound and/or a time of such an actionable sound. The machine learning model may similarly execute the machine learning model using spectrograms of audio data provided by each of the plurality of microphones of the recording device.

The machine learning model may output binary values indicating whether the sets of audio data include audio data of actionable sounds and/or a time of such an actionable sound. If the machine learning model outputs as value indicating none of the sets of audio data include audio data of an actionable sound, the recording device may discard each of the sets of audio data (e.g., remove the sets of audio data from memory) and stop performing the method **2000**. However, if the machine learning model outputs at least one value indicating a set of audio data does include audio data of an actionable sound (e.g., a binary value indicating the set of audio data includes audio data of an actionable sound and/or a timestamp of such an actionable sound), the recording device may determine the set of audio data is associated with an actionable sound. The recording device can identify any other sets of audio data that include values indicating the set of audio data includes audio data of an actionable sound. The recording device can discard any sets of audio data the recording device determines do not include audio of an actionable sound.

In step **2006**, the recording device can determine a location of the actionable sound. The recording device can determine the location of the actionable sound based on the sets of audio data the recording device determined to correspond to (e.g., include audio data of) the actionable sound. The recording device can determine the location of the actionable sound as the location of the actionable sound relative to a camera (e.g., a camera coupled to the housing of the recording device or a camera remote from the recording device). The location of the actionable sound relative to the camera can be a distance and/or direction of the location from the camera. The location can be a or include a vector (e.g., a three-dimensional vector) and/or a distance. To do so, the recording device can identify the times of arrival of the sets of audio data that contain the actionable sound (e.g., times of the actionable sound determined from the different sets of audio data). The recording device can also identify the locations of the different microphones that provided the recording device with sets of data that contain the actionable sound. The recording device can use three-dimensional sound localization techniques based on the locations of the microphones and the times of arrival of the actionable sound. In doing so, the recording device can generate a three-dimensional digital representation (e.g., a three-dimensional vector) of the location (e.g., the source) of the actionable sound relative to the sensors (e.g., identify the location of the actionable sound relative to the microphones). The recording device can determine a vector between the plurality of microphones and the location of the actionable sound. The recording device can identify or determine a vector between the plurality of microphones that provided sets of audio data including the actionable sound (e.g., the middle or centroid of the plurality of microphones) and the camera. The recording device can determine an angle between the two vectors. The recording device can determine the location of the actionable sound relative to the

camera based on the two vectors and the angle such as by using the law of sines or the law of cosines.

In step **2008**, the recording device can rotate a camera. The recording device can rotate the camera for which the recording device determined the relative position of the actionable sound to the camera. The camera can be coupled to the housing of the recording device or be remote from the recording device. The recording device can identify a direction in which the camera is pointing. The recording device can determine an angle between the direction in which the camera is pointing and the location of the actionable sound relative to the camera. Based on the angle, the recording device can determine an azimuth and an elevation in which to rotate the camera to cause the camera to point in the direction of the location of the actionable sound (e.g., cause the location of the actionable sound to be in the field of view of the camera).

Responsive to rotating the camera to point in the direction of the location of the actionable sound, the recording device can activate the camera (e.g., change the state of the camera from “off” to “on”). Upon being activated, the camera can begin recording a video of the field of view of the camera. The camera can transmit the recorded video to a remote computing device. The camera can do so directly across a network or by transmitting the recorded video to the recording device, which can forward the recorded video to the remote computing device. The camera and/or the recording device can do so in real-time to provide a live video feed of the field of view of the camera including the location of the actionable sound. Accordingly, the recording device and camera can operate in tandem to provide the remote computing device with a live feed of the aftermath of an actionable sound.

The recording device can transmit an alert identifying the camera or the recording device to a remote computing device responsive to determining at least one set of audio data corresponds to the actionable sound. The remote computing device can be a computing device of a video management system that operates cameras around municipal area in which the recording device is located. Responsive to receiving the alert, a user accessing the remote computing device can select an option and/or otherwise video a live video feed being provided by the camera. In some cases, the recording device can transmit the determined elevation and azimuth (or pan, tilt, zoom operations) to the remote computing device in the alert. In such cases, the remote computing device can control the camera according to the transmitted elevation and azimuth or pan, tilt, zoom operations. Receipt of the determined elevation and azimuth (or pan, tilt, zoom operations) may cause the remote computing device to automatically control the camera according to the received information. Such may be advantageous, for example, in cases in which the recording device is not in communication with the camera.

FIG. **21** illustrates a sequence **2100** of rotating a camera **2102** of the recording device **1702**, according to an embodiment. In the sequence **2100**, the camera **2102** may initially be in a first state **2104** pointing in a first direction **2106**. The processor of the recording device **1702** may determine a location of a gunshot and/or a direction to point the camera **2102**. The processor may transmit a control signal to the camera **2102** to cause the camera **2102** to rotate to be in a second state **2108** pointing in a second direction **2110** at the location of the gunshot.

FIGS. **22A-B** illustrate different cameras **2202** and **2204** that can be rotated in response to detecting a gunshot, according to an embodiment. The cameras **2202** and **2204**

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can be rotated by a processor controlling the cameras 2202 and 2204 in response to the processor detecting actionable sounds (e.g., gunshots, car crashes, screams, etc.). The processor can be stored remote from or in the housing of the camera 2202 or the camera 2204, in some cases with microphones that continuously record audio of the surrounding environment. The processor can perform the method 2000 to detect actionable sounds from audio data the processor receives from the microphones and rotate one of the cameras 2202 or 2204 to cause the field view of the camera 2202 or 2204 to include the location of the actionable sound. Cameras 2202 and 2204 are example cameras. Any type of camera can be rotated to capture video of a location from which an actionable sound was detected. In some cases, the processor can be remote from the camera 2202 or 2204 and communicate with the camera 2202 or 2204 over a network.

The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skill in the art the steps in the foregoing embodiments may be performed in any order. Words such as “then”, “next”, etc. are not intended to limit the order of the steps; these words are simply used to guide the reader through the description of the methods. Although process flow diagrams may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination may correspond to a return of the function to the calling function or the main function.

The various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

Embodiments implemented in computer software may be implemented in software, firmware, middleware, microcode, hardware description languages, or any combination thereof. A code segment or machine-executable instructions may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc. may be passed, forwarded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, etc.

The actual software code or specialized control hardware used to implement these systems and methods is not limiting of the invention. Thus, the operation and behavior of the systems and methods were described without reference to

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the specific software code being understood that software and control hardware can be designed to implement the systems and methods based on the description herein.

When implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable or processor-readable storage medium. The steps of a method or algorithm disclosed herein may be embodied in a processor-executable software module, which may reside on a computer-readable or processor-readable storage medium. A non-transitory computer-readable or processor-readable media includes both computer storage media and tangible storage media that facilitate transfer of a computer program from one place to another. A non-transitory processor-readable storage media may be any available media that may be accessed by a computer. By way of example, and not limitation, such non-transitory processor-readable media may comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other tangible storage medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer or processor. Disk and disc, as used herein, include compact disk (CD), laser disc, optical disk, digital versatile disk (DVD), floppy disk, and Blu-ray disk where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product.

What is claimed is:

1. A system for detecting gunshots, comprising:
 - a computing device comprising:
 - a device housing configured to removably couple to a light fixture, wherein a photoelectric cell of the light fixture is coupled to an upper surface of the device housing;
 - a microphone inside or mounted to the device housing; and
 - a processor inside the device housing and electrically coupled to the microphone, the processor configured to:
 - receive a set of audio data from the microphone;
 - execute a first machine learning model using a segment of the set of audio data comprising a first plurality of sounds as input to determine whether the segment of the set of audio data is likely to contain a sound of a gunshot; and
 - responsive to determining the segment of the set of audio data is likely to contain the sound of the gunshot based on the execution of the first machine learning model, transmit the segment of the set of audio data to a remote computing device; and
 - the remote computing device comprising:
 - a remote processor configured to:
 - receive the segment of audio data from the computing device responsive to the processor determining the segment of the set of audio data is likely to contain the sound of a gunshot;
 - responsive to receiving a plurality of segments of audio data each comprising a second plurality of sounds and corresponding to the gunshot received from a plurality of computing devices, the plural-

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ity of segments of audio data including the segment of audio data from the computing device, iteratively execute, for each segment of audio data of each computing device, a second machine learning model using the segment of audio data received from the computing device to output a plurality of timestamps of the gunshot for the computing device; and

determine a location of the gunshot based on a device location and only a first timestamp of the plurality of timestamps for each of the plurality of segments of audio data, each first timestamp for each segment of audio data selected based on the first timestamp corresponding to an earliest time of the plurality of timestamps for the segment of audio data.

2. The system of claim 1, wherein the processor is further configured to:

convert the received segment of audio data into a multi-dimensional acoustic envelope sample illustrating one or more sound frequencies of the segment of audio data, and

wherein the processor is configured to execute the first machine learning model using the segment of audio data as input by executing the first machine learning model using the multi-dimensional acoustic envelope sample as input.

3. The system of claim 2, wherein the processor is configured to transmit the segment of audio data to the remote processor by transmitting the multi-dimensional acoustic envelope sample to the remote processor.

4. The system of claim 1, wherein the processor is further configured to:

receive a second segment of audio data from the microphone;

execute the first machine learning model using the second segment of audio data as input to determine the second segment of audio data does not correspond to any gunshots; and

responsive to determining the second segment of audio data does not correspond to any gunshots, discard the second segment of audio data.

5. The system of claim 1, wherein the processor is configured to execute the first machine learning model at predetermined intervals with audio data generated during each of the predetermined intervals.

6. A method, comprising:

receiving, by a processor of an edge recording device, a first set of audio data from a microphone inside or mounted to a housing of the edge recording device, the first set of audio data comprising a sound recording;

executing, by the processor, a first machine learning model using a first segment of the first set of audio data as input to determine the first segment of audio data comprising a first plurality of sounds is likely to contain a sound of a gunshot;

responsive to determining the first segment of the first set of audio data is likely to contain the sound of the gunshot, transmitting, by the processor, the first segment of audio data to a first remote processor;

receiving, by the first remote processor, the first segment of audio data as a segment of audio data of a plurality of segments of audio data received from a plurality of edge recording devices, each of the plurality of segments of audio data transmitted to the first remote processor in response to a determination the segment is likely to contain the sound of the gunshot;

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responsive to receiving the plurality of segments of audio data from the plurality of edge recording devices, iteratively executing, by the first remote processor for each segment of audio data of each edge recording device, a second machine learning model using the segment of audio data received from the edge recording device to output a plurality of timestamps of the gunshot for the edge recording device;

determining a location of the gunshot based on a device location and only a first timestamp of the plurality of timestamps for each of the plurality of segments of audio data, each first timestamp for each segment of audio data selected based on the first timestamp corresponding to an earliest time of the plurality of timestamps for the segment of audio data; and

and

transmitting, by the first remote processor, an indication of the location to a second remote processor of a second remote device.

7. The method of claim 6, further comprising:

converting, by the first remote processor, the first segment of audio data into a spectrogram illustrating one or more sound frequencies of the first segment of audio data,

wherein executing the second machine learning model using the first segment of audio data as input comprises executing, by the first remote processor, the second machine learning model using the spectrogram as input.

8. The method of claim 6, wherein executing the first machine learning model using the first segment of audio data as input comprises executing, by the processor, the first machine learning model to determine whether any gunshots occurred within a time period; and

wherein executing the second machine learning model comprises executing, by the first remote processor, the second machine learning model using a spectrogram generated by the processor or the first remote processor as input to determine which impulses of the spectrogram correspond to gunshots.

9. A system, comprising

a first remote processor of a first remote computing device remote from a set of edge recording devices, the first remote processor coupled to a first remote non-transitory memory of the first remote computing device, wherein the first remote processor is configured to:

receive a segment of audio data comprising a plurality of sounds from each of a subset of the set of edge recording devices, each segment of audio data transmitted to the first remote processor in response to a determination that segment of audio data is likely to contain a sound of a gunshot, the subset comprising a plurality of edge recording devices;

responsive to receiving the segment of audio data from each of the subset of edge recording devices, iteratively execute, for each of the subset of edge recording devices, a machine learning model trained to use the segment of audio data received from the edge recording device to output a plurality of timestamps of the gunshot for the edge recording device;

determine a location of the gunshot based on a device location and only a first timestamp of the plurality of timestamps for each of the plurality of segments of audio data, each first timestamp for each segment of audio data selected based on the first timestamp corresponding to an earliest time of the plurality of timestamps for the segment of audio data;

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and

transmit an indication of the location to a second remote processor of a second remote computing device.

10. The system of claim 9, further comprising a processor of an edge device of the subset of edge devices, the processor in communication with an edge non-transitory memory of the edge device and one or more microphones mounted on or in a housing of the edge device, wherein the processor is configured to:

receive a first set of audio data from the one or more microphones, the first set of audio data comprising a sound recording;

execute a second machine learning model using the first segment of audio data of the set of audio data as input to determine the first segment of audio data is likely to contain the sound of a gunshot, the segment of audio data comprising the plurality of sounds over a defined time period; and

responsive to determining the first segment of audio data is likely to contain the sound of a gunshot, transmit the first segment of audio data to the first remote processor as a segment of audio data from the subset of edge devices.

11. The system of claim 10, wherein the processor is configured to:

convert the first segment of audio data into a spectrogram illustrating one or more sound frequencies of the plurality of sounds of the first segment of audio data, wherein the processor is configured to execute the second machine learning model using the first segment of audio data as input by executing the second machine learning model using the spectrogram as input.

12. The system of claim 11, wherein the processor is configured to transmit the first segment of audio data to the first remote processor by transmitting the spectrogram illustrating the one or more sound frequencies of the first segment of audio data, and

wherein the first remote processor is configured to execute the machine learning model using the first segment of audio data as input by executing the machine learning model using the spectrogram illustrating the one or more sound frequencies of the first segment of audio data.

13. The system of claim 11, wherein the processor is configured to execute the second machine learning model using the spectrogram to determine whether the gunshot occurred within a time period; and

wherein the first remote processor is configured to execute the machine learning model using the spectrogram or a second spectrogram of the first segment of audio data generated by the first remote processor to determine which impulses of the spectrogram or the second spectrogram correspond to the gunshot.

14. The system of claim 9, wherein the first remote processor is configured to:

convert a first segment of audio data of the segments of audio data into a spectrogram illustrating one or more sound frequencies of the first segment of audio data, wherein the first remote processor is configured to execute the machine learning model using the first segment of audio data as input by executing the machine learning model using the spectrogram as input.

15. The system of claim 14, wherein the first remote processor is configured to execute the machine learning model using the spectrogram as input by:

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identifying a plurality of impulses of the spectrogram that each correspond to the gunshot;

identifying a first plurality of timestamps that correspond to the plurality of impulses; and

determining a first time of the gunshot for the first segment of audio data as an impulse with an earliest time stamp of the first plurality of timestamps.

16. The system of claim 9, wherein the first remote processor is further configured to:

execute a firework detection machine learning model to determine a first segment of audio data does not correspond to fireworks,

wherein the first remote processor is configured to execute the machine learning model using the first segment of audio data as input or execute a multilateration model using a first timestamp of the gunshot associated with the first segment of audio data responsive to determining the first segment of audio data does not correspond to fireworks.

17. The system of claim 9, wherein the first remote processor is configured to execute a multilateration model to determine the location of the gunshot by:

generating a list of a plurality of permutations of the subset of edge devices and a potential gunshot location for each of the plurality of permutations;

calculating an average location of the potential location for each of the plurality of permutations;

removing a first permutation from the list responsive to the first permutation having a first potential location farthest from the average location; and

determining the location of the gunshot as the potential location for a permutation remaining on the list after removing the first permutation from the list.

18. The system of claim 9, wherein receipt of the indication of the location by the second remote processor causes the second remote processor to direct an unmanned vehicle to the location.

19. A system comprising:

a computing device comprising:

a device housing configured to removably couple to a light fixture, wherein a photoelectric cell of the light fixture is coupled to an upper surface of the device housing;

a microphone inside or mounted to the device housing; and

a processor inside the device housing and electrically coupled to the microphone, the processor configured to:

receive a set of audio data from the microphone; execute a first machine learning model using a segment of the set of audio data comprising a first plurality of sounds as input to determine whether the segment of the set of audio data is likely to contain a sound of a gunshot; and

responsive to determining the segment of the set of audio data is likely to contain the sound of the gunshot based on the execution of the first machine learning model, transmit the segment of the set of audio data to a remote computing device; and

the remote computing device comprising:

a remote processor configured to:

receive the segment of audio data from the computing device; and

execute a second machine learning model using the segment of audio data received from the comput-

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ing device to output a first timestamp of the
gunshot for the computing device;
responsive to executing the second machine learning
model using the segment of audio data received
from the computing device to output the first
timestamp of the gunshot for the computing
device, query at least one computing device within
a distance threshold of the computing device with
the first timestamp of the gunshot for the comput-
ing device output by the second machine learning
model;
wherein each of the at least one computing device is
configured to:
in response to receiving the query containing the
first timestamp of the gunshot for the comput-
ing device, execute a third machine learning
model using a second segment of audio data
corresponding to the first timestamp of the
gunshot for the computing device to determine
whether the second segment of audio data is
likely to contain the sound of the gunshot; and

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responsive to determining the second segment of
audio data is likely to contain the sound of the
gunshot based on the execution of the third
machine learning model, transmit the second
segment of the set of audio data to the remote
computing device;
wherein the processor of the remote computing
device is configured to:
receive the second segment of audio data from
each of the at least one computing device; and
iteratively execute, for each received second seg-
ment of audio data, a second machine learning
model using the second segment of audio data
to output a second timestamp of the gunshot for
the second segment of audio data that is con-
figured with the first timestamp for input into a
multilateration model to determine a location of
the gunshot.

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